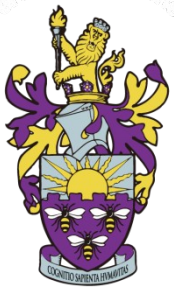
GIS- and Remote-Sensing-Based Landslide Inventory, Susceptibility and Exposure Assessment along the NH-5–Satluj River Corridor from Kafaur to Khab, Kinnaur District, Himachal Pradesh, India

A dissertation submitted to the University of Manchester for the degree of MSc Geographical Information Science in the Faculty of Humanities



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Abstract

Landslides are a recurring hazard in the steep and highly dissected terrain of Kinnaur, where slope failures can affect transport routes, land resources and nearby settlements along the Satluj Valley. This study assessed the spatial distribution of mapped landslides, relative landslide susceptibility and associated exposure, with particular focus on the NH-5–Satluj River corridor from Kafaur to Khab.

A landslide inventory was developed through manual interpretation of high-resolution Google imagery from November 2025. Road-proximity patterns were examined within a 1000 m analytical corridor using three non-overlapping distance bands, while Sentinel-2 land-use/land-cover data and village point locations were used for exposure analysis. Relative susceptibility across wider Kinnaur was assessed using an eight-factor GIS Weighted Overlay, with a separate Kriging interpolation used only to explore the spatial pattern of mapped landslide area.

A total of 179 landslides were mapped within the corridor, of which 102 (57.0%) occurred within 0–300 m of NH-5, decreasing to 44 (24.6%) within 300–600 m and 33 (18.4%) within 600–1000 m. Mapped landslides overlapped 6.3026 km² of LULC, dominated by rangeland (58.09%) and bare ground (38.10%), while 91 village point locations were identified within the analytical corridor. Across wider Kinnaur, Medium susceptibility formed the largest class (44.60%), followed by Low (26.67%), High (26.09%) and Very High (2.64%).

The results provide a spatial basis for prioritising field inspection and more detailed slope assessment, while representing relative susceptibility and exposure rather than confirmed damage, quantified risk or proven road causation.

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Finally, I would like to thank my family and friends for their patience, encouragement and support throughout my time at Manchester. Completing this degree has been both challenging and rewarding, and I am deeply grateful to everyone who supported me along the way.

Declaration

No portion of the work referred to in the dissertation has been submitted in support of an application for another degree or qualification of this or any other university or other institute of learning.

Statement of AI use

In the process of completing this submitted work, the following artificial intelligence (AI) tools were used:

- ChatGPT (OpenAI) was used to help explain selected concepts in simpler language, support research planning and organisation, prioritise stages of work, and provide feedback on the structure and clarity of selected sections during revision and to improve vocabulary.
- Google NotebookLM was used to support the understanding of selected academic papers uploaded to the platform, clarify concepts within the literature, and assist in locating relevant information within those sources.
- Grammarly was used for grammar, sentence clarity and general language checking during proofreading and revision.
- Scite (Research Solutions) was used for citation analysis and to help identify potentially relevant papers, including literature supporting, contrasting or citing key studies. Original publications were consulted before any source was used or cited.

- ResearchRabbit was used to explore citation networks between key papers, including references, cited-by relationships and related literature, to support literature discovery.
- Elicit was used to support focused literature searches, identify potentially relevant academic studies and compare key findings across selected publications.

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1. Introduction

1.1 Background and Study Significance

Landslides are among the major geomorphological hazards affecting mountainous regions and can result in loss of life, damage to property, disruption of transport networks, degradation of agricultural and natural land, and damage to other infrastructure. The Indian Himalayan region is particularly susceptible to slope instability because of its steep and highly dissected terrain, complex geological structure, active geomorphic processes and strong seasonal climatic influence. Reliable identification and spatial assessment of landslides are therefore important for understanding their distribution and for supporting disaster management, land-use planning and infrastructure protection. Gupta and Sah (2008a) similarly emphasised that the preparation of a reliable landslide inventory is an essential basis for analysing the relationship between landslide occurrence and environmental conditions.

Himachal Pradesh is an almost wholly mountainous state, with elevations ranging from approximately 350 to 6,975 m above mean sea level. Its strongly dissected relief, deep valleys and diverse geological and climatic conditions contribute to considerable spatial variation in slope stability. The Himachal Pradesh State Disaster Management Authority (HPSDMA) identifies landslides as a recurring hazard in the state and recognises Kinnaur as one of the districts with high landslide vulnerability. The state-level assessment also identifies hydro-meteorological conditions, fragile geological strata and human modification associated with activities such as road construction, tunnelling and hydropower development as factors that can aggravate slope instability (HPSDMA, 2012).

Within Kinnaur, the Satluj Valley represents a particularly important setting for landslide investigation. Previous research documents a long history of mass movements in the valley and demonstrates that steep slopes, high relief, geological discontinuities and underlying lithology interact with anthropogenic disturbances to influence slope failure (Gupta and Sah, 2008a). Their study further documented increasing development pressure associated with hydropower construction, road excavation and widening, and changing land use. The principal highway follows the deeply incised Satluj Valley and provides an important

transport connection for settlements distributed along the corridor, creating a close spatial relationship between unstable hill slopes, infrastructure and human activity.

Individual landslides illustrate the significance of this interaction. The Urni landslide developed into a complex slope failure that partially dammed the Satluj River and damaged approximately 200 m of NH-05, while the crown of the landslide lies close to Urni village (Kumar et al., 2019). Such events demonstrate that landslides within the valley can influence not only hill slope stability but also river processes, road connectivity and nearby settlements.

Remote sensing and Geographic Information Systems (GIS) provide practical tools for investigating these relationships across steep and difficult-to-access Himalayan terrain. Recent studies in Kinnaur have used GIS to integrate landslide inventories with factors including slope, aspect, lithology, soil, drainage, rainfall and land use/land cover for susceptibility assessment (Qazi et al., 2023), while Sentinel-1 PS-InSAR has been applied to detect surface deformation and potential landslide areas in the district (Joshi et al., 2024). Against this background, the present dissertation focuses on the NH-5–Satluj River corridor from Kafaur to Khab, integrating landslide inventory mapping, relative susceptibility assessment and spatial exposure analysis using GIS and remote-sensing techniques.

1.2 Research Focus and Scope

The geographical focus of this dissertation is the NH-5–Satluj River corridor from Kafaur to Khab in Kinnaur district, Himachal Pradesh, within the western Himalaya. The corridor follows the deeply incised Satluj Valley through steep mountainous terrain and provides an important spatial link between settlements, land-use resources and transport infrastructure. Although the wider Kinnaur district provides the environmental context for the susceptibility analysis, the principal applied focus of the research is the relationship between landslide occurrence and exposure along this road corridor.

The study is organised around three interconnected components: landslide inventory, relative susceptibility and exposure. Landslide features interpreted from available high-resolution imagery are used to establish their spatial distribution, while terrain, geological, hydrological, climatic and land-cover information are integrated within GIS to examine

relative susceptibility. The exposure component considers the spatial relationship of mapped landslides with NH-5, settlements and land-cover resources, including agricultural and built-up areas. This retains the physical and socio-economic dimension of the research without assuming that spatial proximity necessarily represents confirmed damage.

For the road-corridor analysis, a 1000 m analytical extent is used around NH-5, with the detailed distance-band analysis described in the Methodology chapter. This boundary provides a consistent spatial framework for examining road-proximal landslide occurrence and exposure; it is not interpreted as a maximum physical distance over which landslides or associated cascading processes may generate impacts. The corridor analysis therefore complements, rather than replaces, the wider district-scale assessment by providing a more focused examination of the geographical relationships between mapped landslides, the highway and exposed elements along the Satluj Valley.

1.3 Aim and Research Questions

This study aims to assess the spatial distribution of mapped landslides, relative landslide susceptibility and associated exposure in Kinnaur District, with particular focus on the NH-5–Satluj River corridor from Kafaur to Khab, using GIS and remote-sensing techniques. In doing so, the following research questions will be answered:

1. How are mapped landslide occurrences spatially distributed in relation to the NH-5–Satluj River corridor between Kafaur and Khab?
2. What spatial patterns of relative landslide susceptibility emerge from the integration of the selected terrain and environmental conditioning factors?
3. Which land-cover resources spatially overlap mapped landslides, which village point locations occur within the 1000 m NH-5 analytical corridor, and what are the implications of these spatial patterns for landslide management?

2. Literature Review

2.1 Landslide Processes and the Satluj–Kinnaur Context

Landslide occurrence in the Indian Himalaya reflects the interaction of several physical and environmental factors rather than a single controlling condition. More generally, Varnes

(1978) explains slope failure through the interaction of factors that increase shear stress and those associated with low or reduced shear strength, emphasising that instability commonly develops through several interacting conditions rather than a single factor. The region is characterised by steep and highly dissected terrain, complex and locally weak lithology, active geological structures and considerable spatial variation in precipitation. These conditions are further influenced by land-cover change and modification of hill slopes associated with roads and other development. In their review of rainfall-induced landslides in the Indian Himalayan region, Dikshit et al. (2020) identify steep relief, fragile lithology, geological complexity and rainfall as important components of Himalayan slope instability, while also recognising the increasing significance of anthropogenic modification.

These relationships are particularly relevant in the Satluj Valley, where the river has produced a deeply incised mountainous landscape and the principal transport corridor occupies steep valley-side terrain. Gupta and Sah (2008a), through their analysis of mass movements along a 130 km section of the Satluj Valley, demonstrated considerable spatial variability in landslide activity and related this variability to geological and geomorphological conditions as well as increasing human pressure. Their study particularly highlighted road excavation and widening, blasting associated with infrastructure development, hydropower construction and changing land use as activities capable of disturbing already sensitive slopes. The importance of the road–slope relationship is therefore not limited to the presence of naturally unstable terrain, but also concerns the way in which development modifies that terrain.

Regional GIS-based studies provide further evidence that landslide susceptibility in Kinnaur needs to be considered through a combination of environmental variables. Jamwal et al. (2020) assessed landscape vulnerability in the upper Satluj basin using parameters including slope, slope aspect, soil texture, lithology, precipitation, land use/land cover, river morphometry and anthropogenic activity. Although their study addressed wider landscape vulnerability rather than the specific NH-5 exposure framework used in this dissertation, it demonstrates the relevance of integrating multiple physical factors in a GIS environment. Similar regional work by Kuniyal et al. (2019) also emphasised the interaction between geomorphological conditions and development pressures associated with hydropower projects in the Satluj catchment.

Individual landslides further demonstrate how these factors operate together at a local scale. The Urni landslide is an important example because it developed as a complex slope failure on the right bank of the Satluj River, partially dammed the river and damaged approximately 200 m of NH-05 (Kumar et al., 2019). Such cases illustrate that slope failure in Kinnaur can have interconnected effects on the physical landscape, transport infrastructure and nearby settlements.

Overall, previous research shows that landslide activity in the Satluj–Kinnaur region results from the combined influence of terrain, geology, precipitation, drainage and surface conditions, while road and hydropower development can provide additional disturbance to susceptible slopes. These findings provide the physical basis for examining slope, elevation, aspect, geology, soil texture, precipitation, stream characteristics and land use land cover within the relative susceptibility assessment undertaken in this dissertation.

2.2 GIS and Remote-Sensing Approaches to Landslide Inventory, Susceptibility and Exposure

Remote sensing and Geographic Information Systems (GIS) are widely used for landslide investigation, particularly in mountainous regions where field access can be difficult and time-consuming. High-resolution and satellite imagery allow landslide features to be identified and mapped over large and inaccessible areas, while GIS provides a common environment for relating mapped landslides to terrain, environmental and land-use information. At the national scale, Martha et al. (2021) demonstrated the use of satellite imagery for developing a polygon-based landslide inventory across the major landslide-prone regions of India and subsequently considered its relationship with terrain and socio-economic exposure. At a local scale in Kinnaur, Kumar et al. (2019) used multi-temporal high-resolution Google Earth imagery to examine the development of the Urni landslide. Such studies demonstrate the value of remotely sensed imagery for landslide inventory development, although interpretation can be influenced by image availability, visibility of landslide features and the temporal coverage of available imagery.

GIS also allows landslide susceptibility to be examined by integrating several terrain and environmental conditioning factors. In the upper Satluj basin, Jamwal et al. (2020) combined topographic, geological, soil, land-use and other environmental information within a GIS-

based vulnerability assessment. Jamwal and Sharma (2022) similarly used a geospatial overlay approach incorporating terrain and environmental parameters for landslide assessment in Kinnaur. These studies demonstrate the practical value of GIS where different spatial datasets need to be combined and interpreted within a common analytical framework.

Other studies have used more statistically derived approaches to susceptibility mapping. Qazi et al. (2023), for example, assessed landslide susceptibility in Kinnaur District using Frequency Ratio, Information Value and Weight of Evidence models. Frequency Ratio and logistic regression approaches have also been applied in other regions (Lee and Pradhan, 2007; Chen et al., 2015). These provide useful methodological comparisons because the contribution of conditioning factors is derived more directly from their spatial relationship with observed landslides, whereas weighted or index-based approaches depend more strongly on the selection, classification and weighting of the chosen factors. These methods therefore represent different approaches to susceptibility assessment and should not be treated as directly equivalent.

Remote sensing can also be used to examine slope movement through time rather than only mapping visible landslide features. In Kinnaur, Joshi et al. (2024) applied Sentinel-1 PS-InSAR to investigate surface deformation at selected locations between 2016 and 2021. This differs from imagery-based landslide inventory mapping because repeated radar observations are used to identify ground displacement through time. It therefore provides an example of active deformation monitoring rather than the inventory-based approach adopted in the present dissertation.

A road-corridor perspective is particularly relevant where landslides interact closely with transport infrastructure and settlements. Sur et al. (2020) used earth-observation data and GIS within a fuzzy AHP susceptibility assessment of the Kalsi–Chakrata road corridor in the Lesser Himalaya, incorporating road proximity alongside terrain, geological, rainfall and land-cover factors. The study also considered villages in relation to the susceptibility pattern, illustrating how corridor-based analysis can connect landslide conditions with elements potentially exposed along a transport route. At a broader scale, Martha et al. (2021) similarly showed how landslide inventories can be considered alongside population and infrastructure information when examining exposure.

Taken together, previous studies demonstrate three related applications of GIS and remote sensing: landslide inventory mapping, susceptibility assessment and spatial exposure analysis. These provide the methodological context for this dissertation. The primary focus here is the Kafaur–Khab NH-5 corridor, where mapped landslide occurrence, road-distance relationships, land-cover resources and settlements are examined. A GIS-based relative susceptibility assessment for the wider Kinnaur study region provides the broader environmental context for this corridor-focused analysis. Statistical susceptibility models and deformation-monitoring techniques discussed above are therefore used only as methodological comparisons and were not applied in the present study.

2.3 Synthesis of Previous Research and Research Gap

Previous research has established a strong understanding of landslides in Kinnaur and the Satluj Valley through regional inventories, site-specific investigations and wider susceptibility assessments. Gupta and Sah (2008a) examined the spatial distribution of mass movements along the Satluj Valley and highlighted the influence of geomorphic conditions and human modification of slopes. Individual landslides such as Urni have also been investigated in greater detail using high-resolution imagery and geotechnical analysis (Kumar et al., 2019), while more recent research has applied PS-InSAR to examine active ground deformation at selected locations in Kinnaur (Joshi et al., 2024).

GIS-based studies have also assessed landscape vulnerability and landslide susceptibility across Kinnaur and the upper Satluj basin by integrating terrain, geological, climatic and land-use information (Jamwal et al., 2020; Jamwal and Sharma, 2022; Qazi et al., 2023). At a wider scale, Martha et al. (2021) demonstrated how landslide inventories can also be related to population and infrastructure exposure. Road-corridor studies elsewhere in the Indian Himalaya have similarly shown the value of examining landslide susceptibility in relation to transport infrastructure and nearby settlements (Sur et al., 2020).

However, within the Satluj–Kinnaur context, comparatively less attention has been given to a focused assessment of landslide occurrence, road-distance relationships and land-cover and settlement exposure along a defined section of NH-5, while also considering the wider spatial pattern of relative landslide susceptibility across Kinnaur. The present dissertation therefore addresses this applied integration gap by combining a corridor-focused landslide

inventory and exposure assessment for the Kafaur–Khab section with a broader GIS-based susceptibility assessment of Kinnaur District. This approach complements, rather than replaces, the regional and site-specific studies already available.

3. Methods

3.1 Study Area, Data and Analytical Design

The study is located in Kinnaur District, Himachal Pradesh, in the western Himalaya. The district is characterised by a strongly dissected mountain landscape ranging from sub-humid to dry-temperate conditions, with steep valley sides, considerable topographic variation and complex geological and lithological characteristics. The principal applied study area follows the NH-5–Satluj River corridor from Kafaur in the west (31°34′37.63″ N, 77°50′28.835″ E) to Khab in the east (31°48′23.574″ N, 78°38′43.316″ E). NH-5 follows the Satluj Valley and provides an important transport connection between settlements and land-use areas distributed along the mountainous valley system. The location and spatial extent of the study corridor are shown in Figure 1.

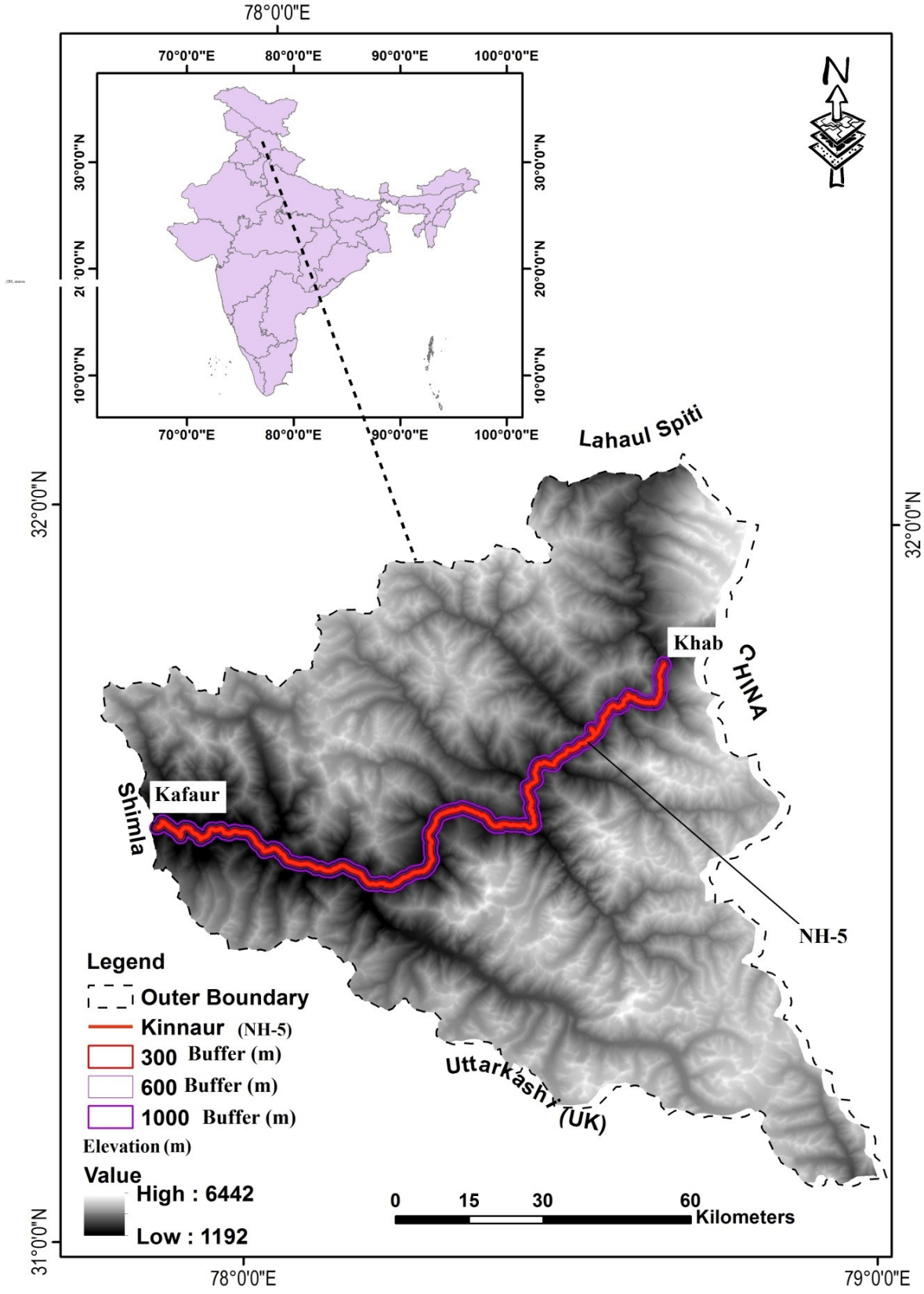


Figure 1 Location of the Kafaur–Khab NH-5–Satluj corridor within Kinnaur District.

The analytical design operates at two related spatial scales. The principal corridor-scale component examines mapped landslide occurrence and exposure within a 1000 m analytical extent around NH-5, while the wider Kinnaur study region is used for the relative susceptibility assessment. The corridor is subsequently analysed using the three non-overlapping road-distance bands described in Section 3.2. This two-scale design allows detailed road-proximal relationships to be examined while retaining a broader regional representation of environmental susceptibility.

The spatial database combines remotely sensed imagery with terrain and secondary thematic GIS datasets. Landslide features were interpreted from high-resolution Google imagery dated November 2025 and processed as vector/KML features. Land-use/land-cover information was obtained from the Esri/Impact Observatory Sentinel-2 10 m Land Use/Land Cover product (Karra et al., 2021). The specific annual product version used in the project was not retained in the available metadata. Terrain information was derived from a 12.5 m ALOS DEM, while geological information was obtained from Geological Survey of India mapping at 1:50,000 scale and soil-texture information from National Bureau of Soil Survey and Land Use Planning mapping at 1:50,000. Precipitation was represented using annual precipitation for 2025 derived from daily CHRS PERSIANN-CDR data at 0.25° spatial resolution (Ashouri et al., 2015).

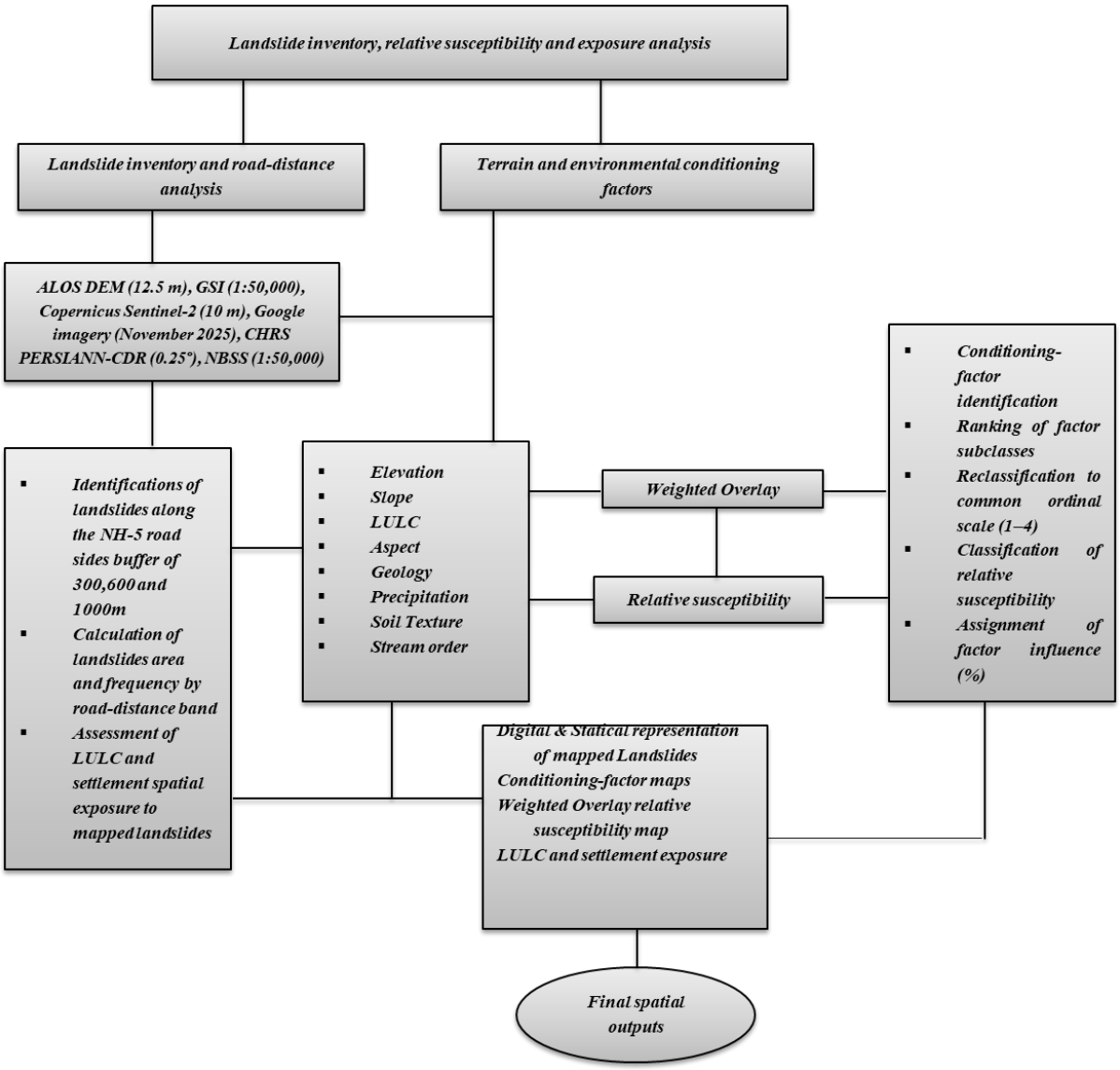


Figure 2 Data sources and analytical workflow

3.2 Landslide Inventory, Road-Distance and Exposure Analysis

Because comprehensive field mapping across the steep and difficult-to-access Satluj Valley was not practicable, the corridor landslide inventory was developed primarily through manual interpretation of high-resolution Google Earth imagery dated November 2025. Visible landslide boundaries were digitised as vector features, initially recorded in KML format and subsequently processed within the GIS environment. Landslide polygons were used to calculate individual mapped areas and examine their spatial distribution along the Kafaur–Khab section of NH-5. For inventory reporting, the mapped landslides were grouped using polygon-area thresholds: Small (<0.01 km²), Medium (0.01 to <0.10 km²) and Large (≥0.10 km²). The observed minimum and maximum areas within each resulting class are

reported in Appendices 1–3. Photographs taken by the author on 15 April 2025 were used only as supplementary visual context for selected landslide locations. They were not collected as part of the formal research design and were not treated as systematic validation of the inventory. Similar use of high-resolution imagery for landslide delineation has previously been applied in Kinnaur (Kumar et al., 2019).

The NH-5 road layer was derived from Survey of India 1:50,000 Open Series Map data (Survey of India, 2025) and used to generate a 1000 m analytical corridor. Road-distance and area operations were undertaken using the projected WGS 1984 / UTM zone 44N coordinate reference system (EPSG:32644), with metres as the linear unit and square kilometres used for reporting area. The corridor was divided into three non-overlapping distance bands: 0–300 m, 300–600 m and 600–1000 m. The final inventory counts were mutually exclusive between the bands, meaning that each mapped landslide was counted once. However, the precise assignment rule applied to any landslide footprint crossing a band boundary was not retained in the available analytical record. Raster-based mapped landslide area was summarised separately within each distance band, using the procedure described below. This provided a consistent spatial basis for comparing landslide occurrence with increasing distance from NH-5 rather than demonstrating a causal effect of road construction on slope failure.

For the exposure analysis, the mapped landslide polygons were converted to raster format using the ArcGIS Polygon to Raster tool with an output cell size of 10 m. The resulting landslide raster was compared spatially with the 10 m Sentinel-2 LULC raster within the same three road-distance zones. For each zone and land-cover class, raster cells representing mapped landslide area were counted. As each 10 m × 10 m cell represented 100 m^2, raster area was converted to square kilometres using:

$$\textbf{Raster area } (km^2) = \text{number of cells} \times (10\ m \times 10\ m) \div 1{,}000{,}000$$

This is equivalent to multiplying the number of cells by 0.0001 km^2. The summed raster-based landslide area within each distance band provided the denominator used to calculate the proportional LULC composition reported in Table 2. Detailed processing settings, including the Polygon to Raster cell-assignment and priority settings, snap raster and exact grid alignment, were not retained and are therefore not reconstructed

Settlement exposure was examined separately from the LULC area calculation. The village-location records were obtained from the Survey of India Administrative Boundary Database and represented as geographic point coordinates (Survey of India, 2025). A total of 91 village points were located within the 1000 m analytical corridor and mapped in relation to NH-5 and the landslide inventory. These points therefore indicate potential spatial exposure and should not be interpreted as village-boundary polygons directly intersected by landslide footprints or as confirmation of damage to individual buildings.

A separate wider-Kinnaur inventory and LULC comparison already present in the GIS dataset was retained only to provide district-scale context. Because the source and mapping history of the wider inventory were less clearly documented than those of the manually mapped corridor inventory, it was not treated as directly equivalent to the corridor dataset. The corridor inventory therefore remains the principal dataset used for the road-distance and exposure analyses.

3.3 Relative Susceptibility and Spatial Interpolation

Relative landslide susceptibility was assessed across the wider Kinnaur study region using eight conditioning factors representing topographic, surface, geological, climatic and hydrological conditions: slope, elevation, land use/land cover (LULC), aspect, geology, soil texture, precipitation and stream order. The data sources and resolutions for these layers are introduced in Section 3.1. Within the susceptibility workflow, slope, elevation and aspect were derived from the 12.5 m ALOS DEM, while the drainage network was generated from the same DEM through sink filling, flow-direction analysis, stream definition and segmentation, followed by stream ordering. Soil-texture polygons were converted to raster format before their inclusion in the overlay. The resulting conditioning-factor layers are presented in Figures 3–10.

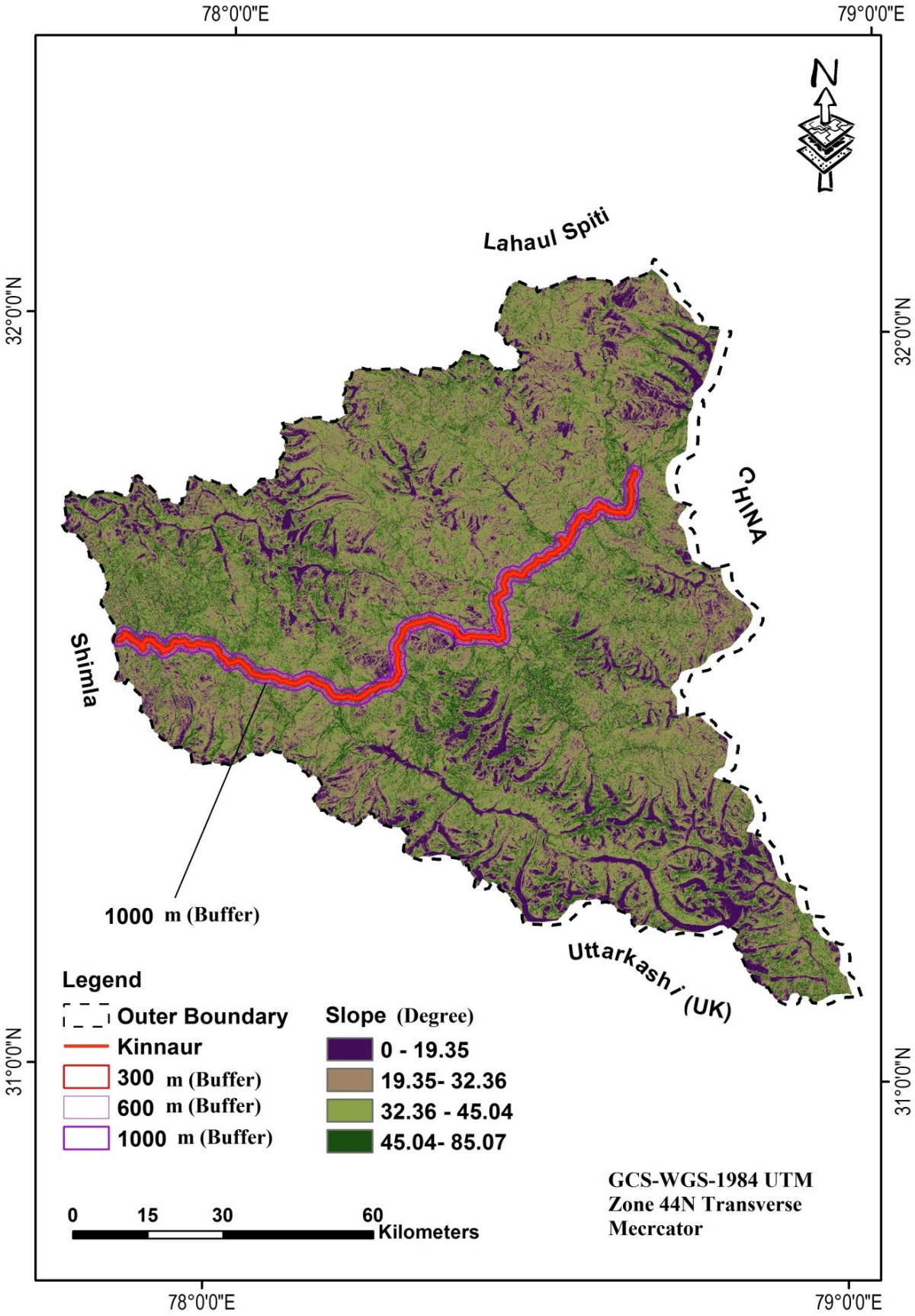


Figure 3 Slope of the Kinnaur study region

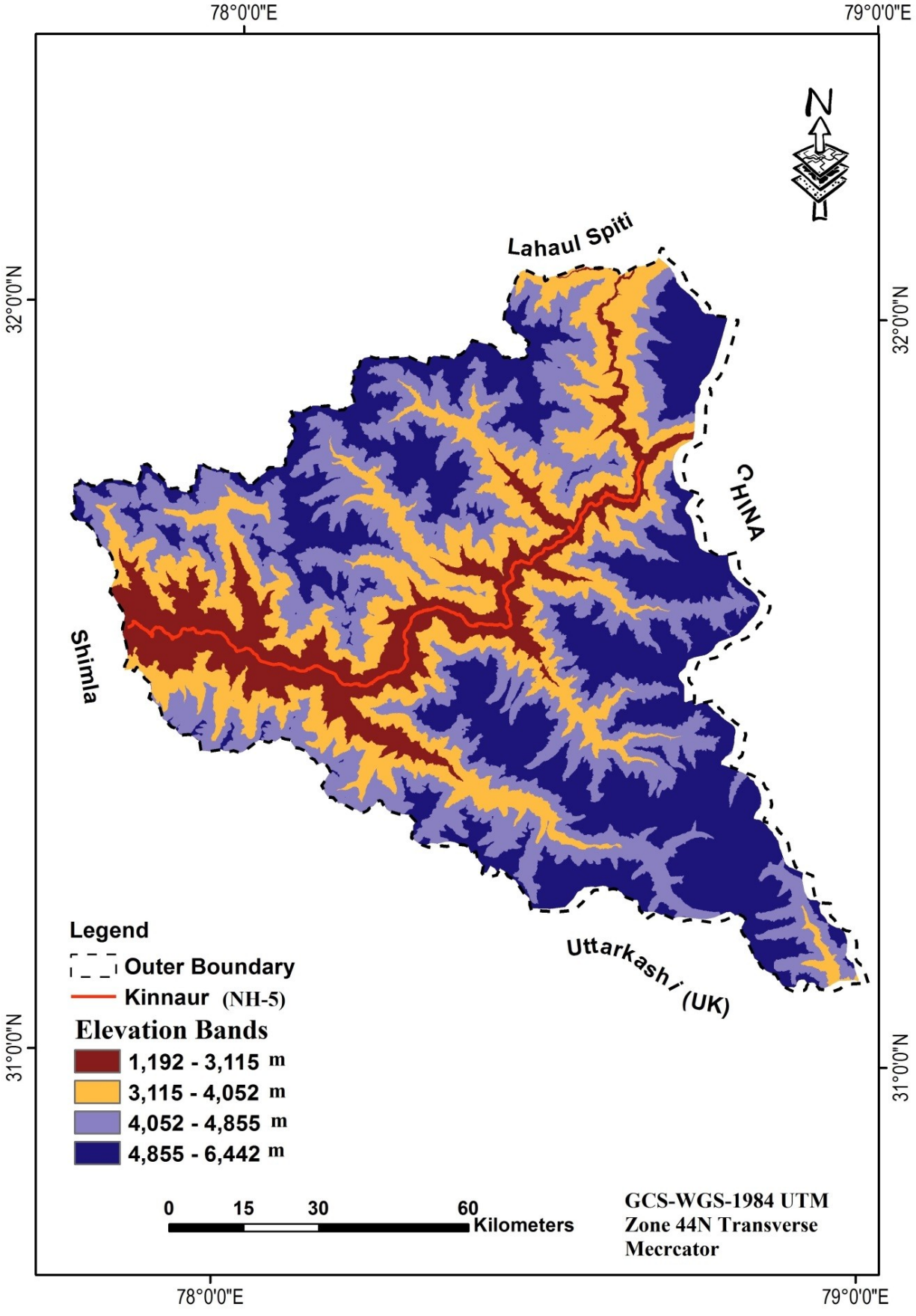


Figure 4 Elevation of the study region

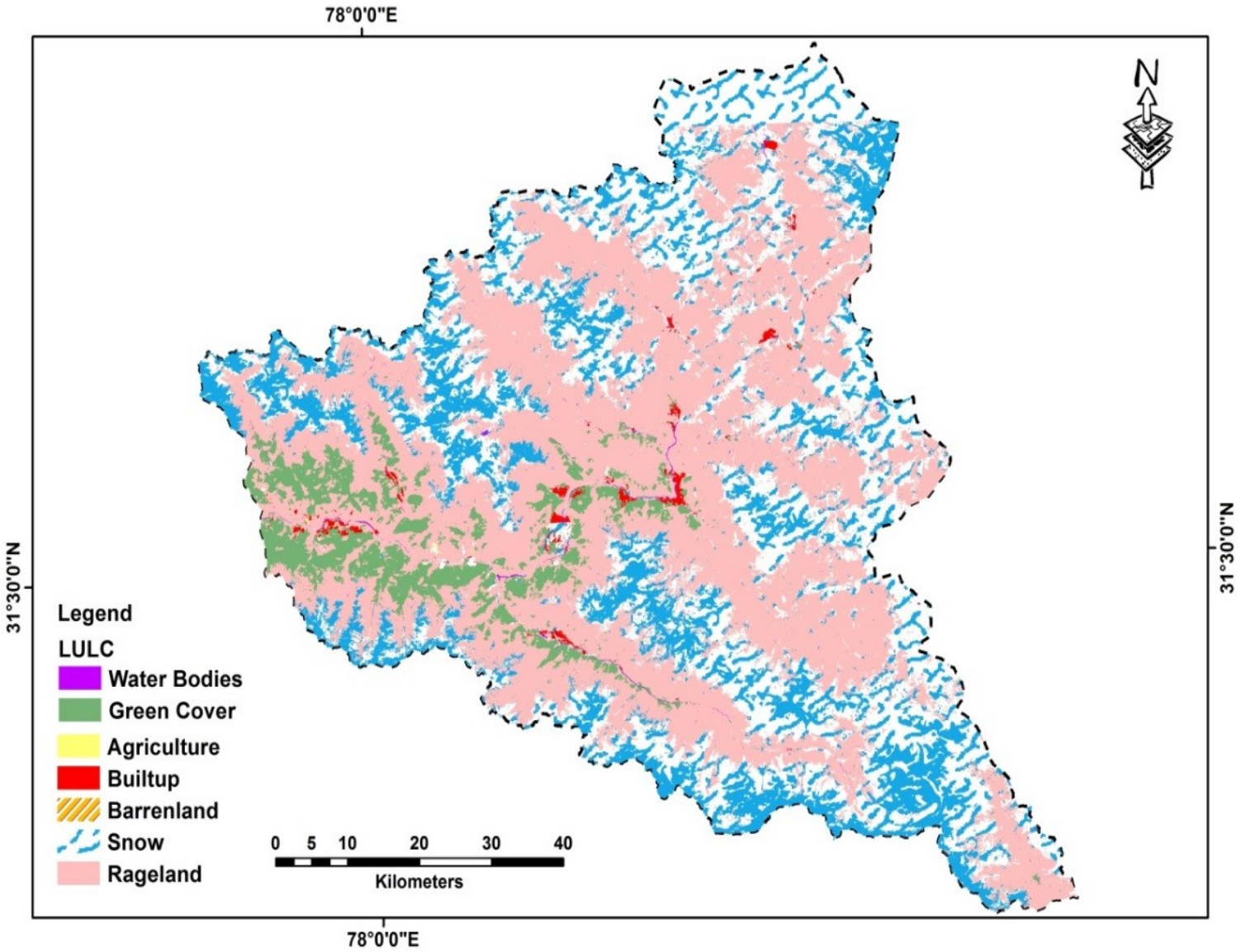


Figure 5 Land Use Land Cover (LULC) status of study region

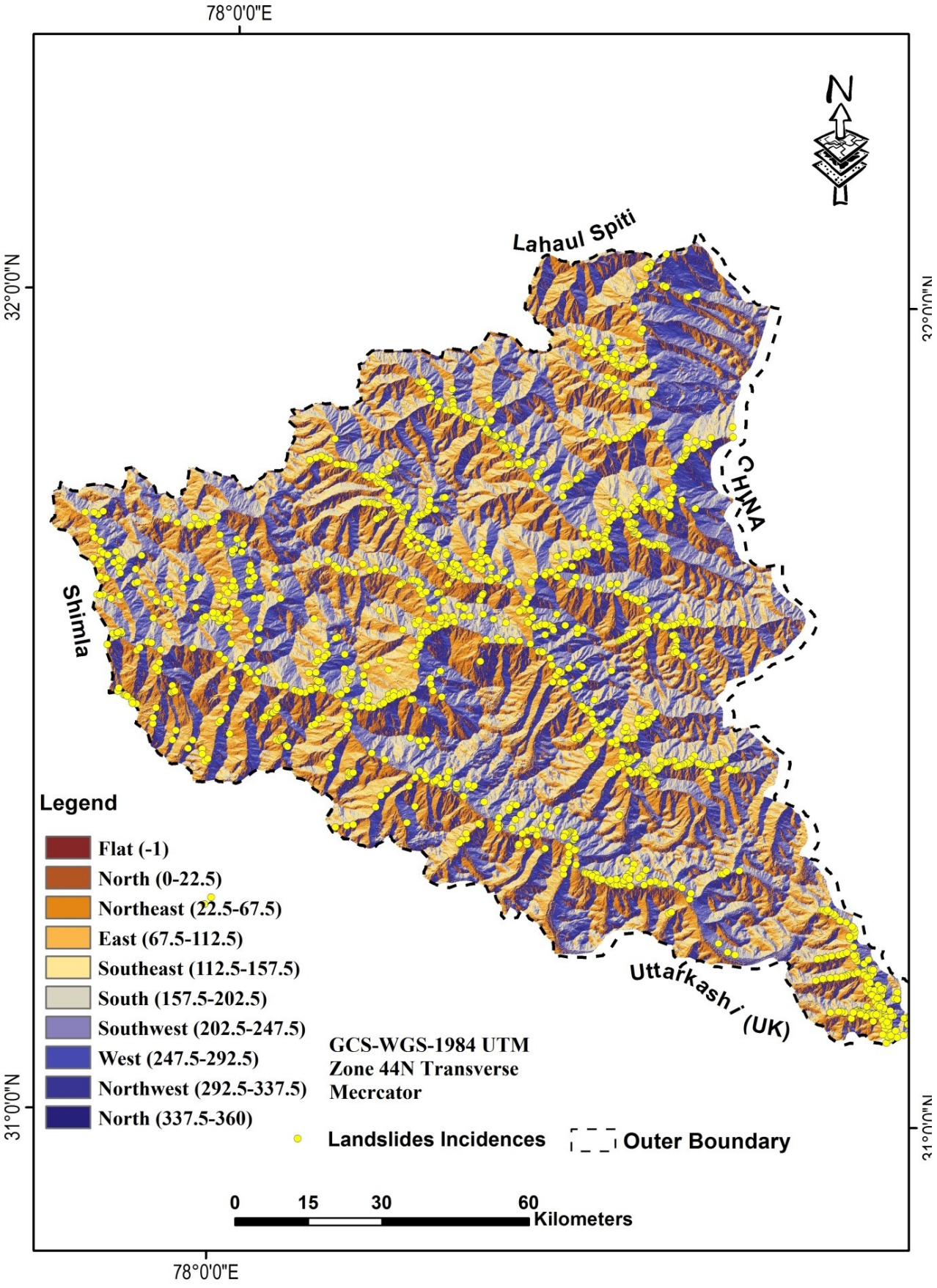


Figure 6 Slope aspect of the study region

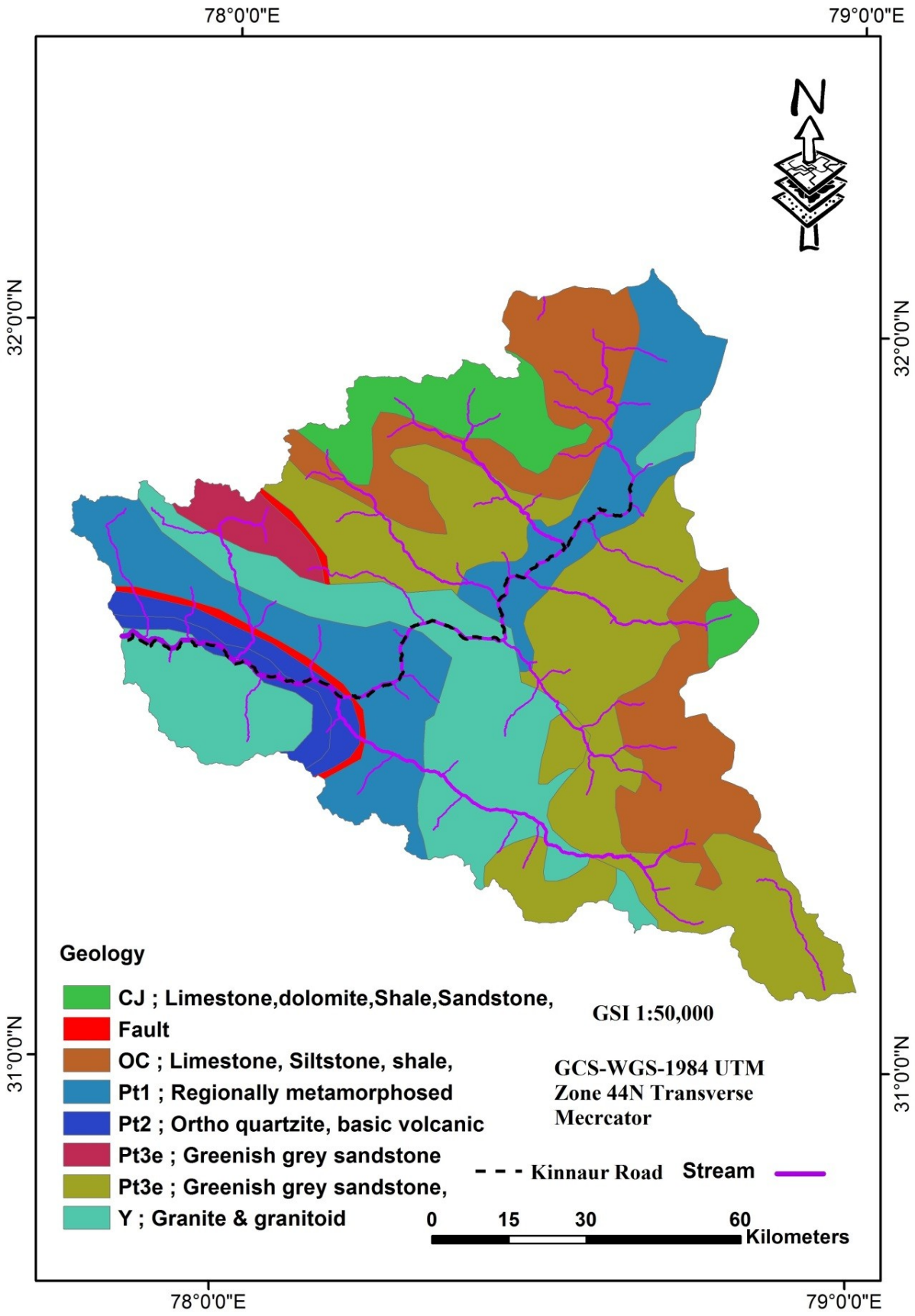


Figure 7 Geology of study region

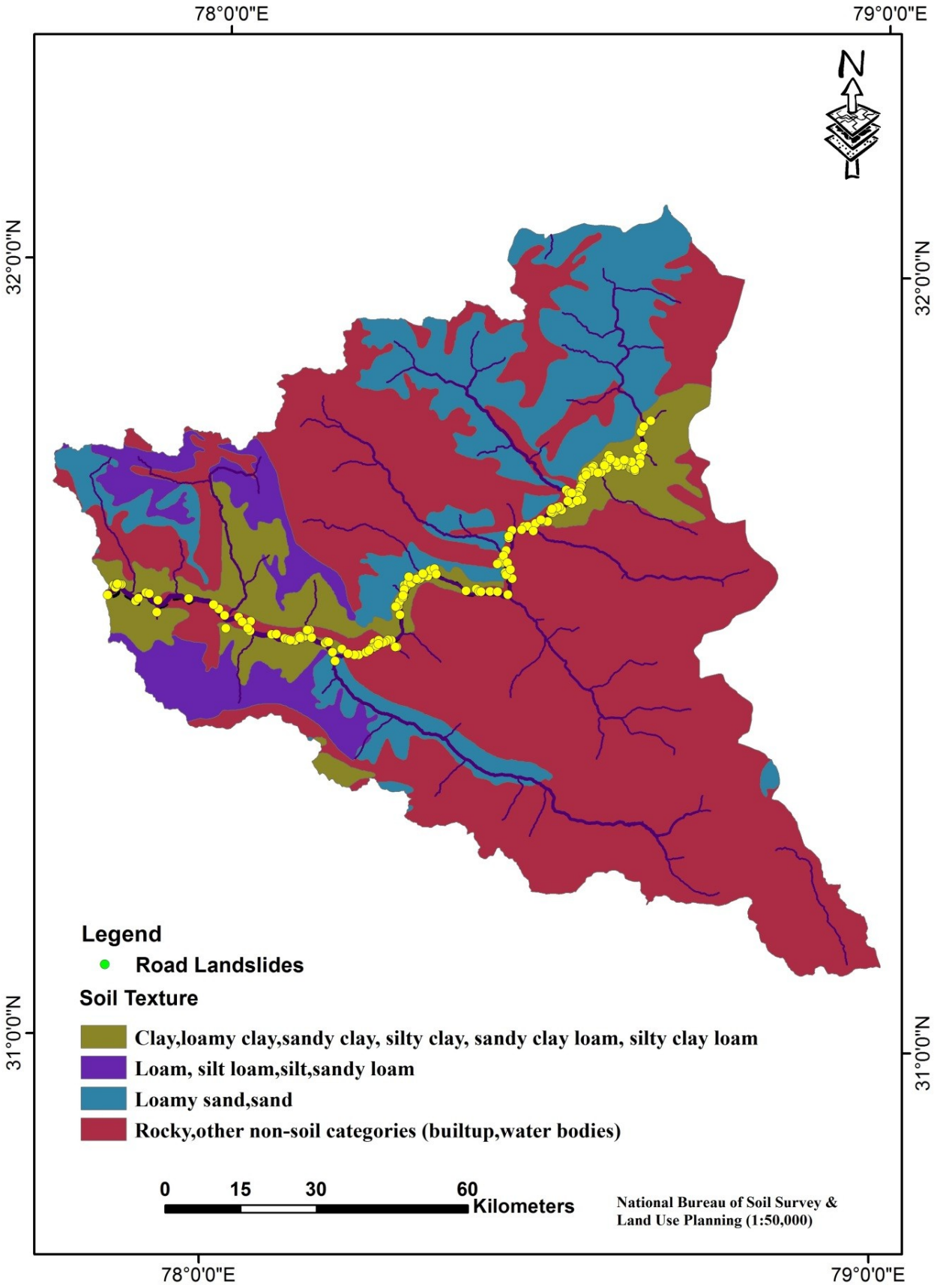


Figure 8 Soil Texture of study region

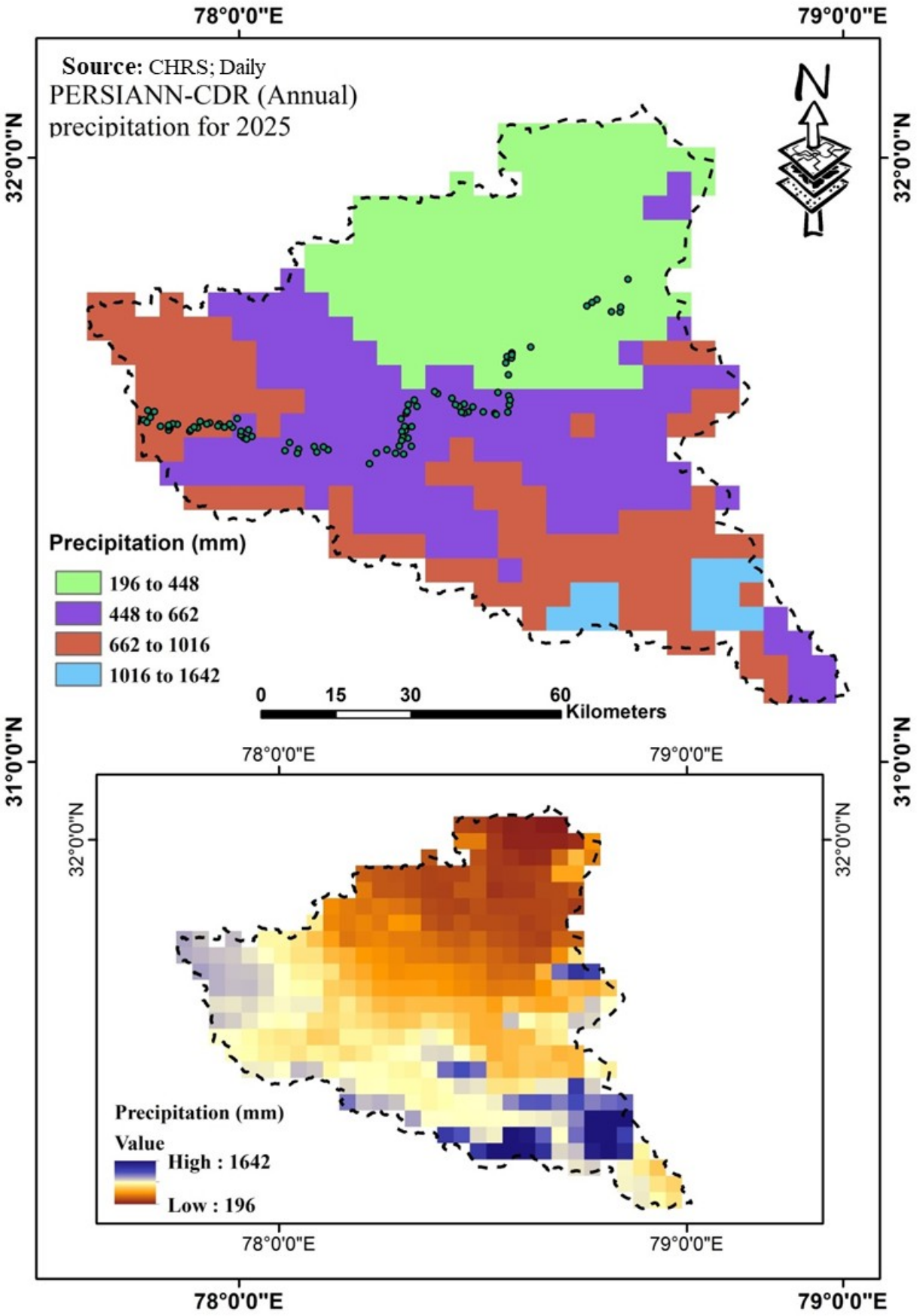


Figure 9 Precipitation across study region

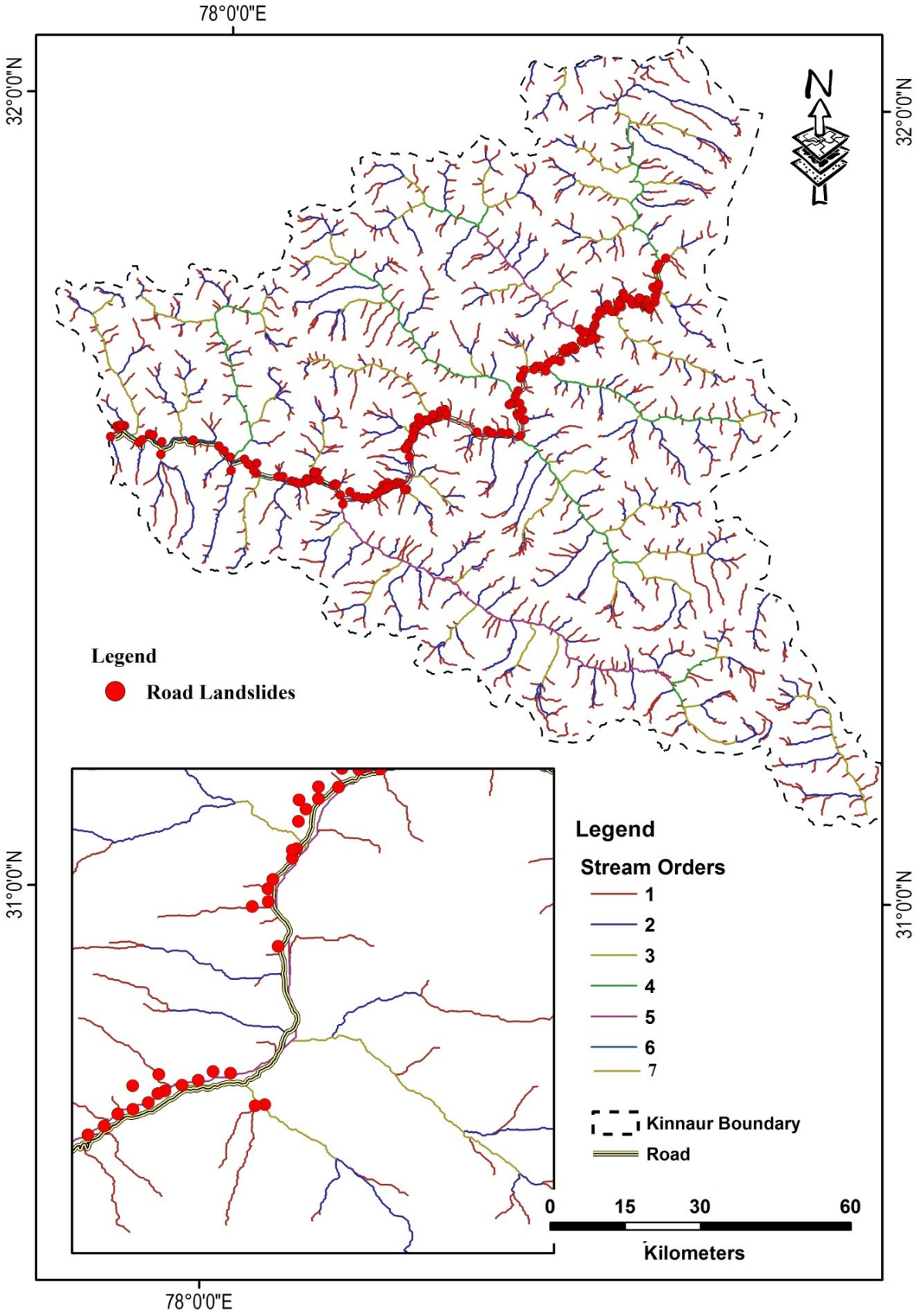


Figure 10 Stream-order network of the study region

Each factor was subdivided into classes and reclassified onto a common ordinal evaluation scale from 1 to 4. Rank 1 represented comparatively lower relative susceptibility and rank 4 comparatively higher relative susceptibility. For presentation in Table 1, these integer ranks are also shown as normalised equivalents of 0.25, 0.50, 0.75 and 1.00. The integer 1–4 ranks were used as the evaluation scale within the Weighted Overlay; the normalised values are therefore descriptive equivalents rather than fuzzy-membership scores. The complete subclass boundaries and their assigned ranks are provided in Table 1 rather than repeated individually in the text.

Factor influence was assigned using a knowledge-driven weighting scheme based on study-specific expert-informed judgement of the relative importance of the selected conditioning factors. The general use of expert knowledge to assign thematic weights and subclass ranks in Himalayan landslide susceptibility mapping is established by Pathak (2016), while regional studies in Kinnaur and the Satluj basin provided additional context for the importance of the selected factors (Jamwal et al., 2020; Jamwal and Sharma, 2022). The specific ranks and percentage influences used here were assigned for the present analysis and were not taken directly from these studies. Slope received the largest influence at 35%, elevation and precipitation 15% each, geology and LULC 10% each, and aspect, soil texture and stream order 5% each. No AHP pairwise-comparison matrix, consistency-ratio calculation or fuzzy-membership analysis was undertaken, and no independent statistical calibration was performed.

Table 1 Susceptibility factor classes, ranks and weights

S.No.	Parameters (m)	Classification of parameters	Area (km^2)	Area(%)	Rank	Normalised score	Susceptibility	Weighted Overlay
1.	**Slope**	0 ° to 19.35 °	1083.1	17.191	1	0.25	L	**35**
		19.35 ° to 32.36 °	2033.3	32.272	2	0.50	M	
		32.36 ° to 45.04 °	2189	34.744	3	0.75	H	
		45.04 ° -85.07°	994.98	15.792	4	1.00	VH	
2.	**Elevation**	1192 to 3115m	772.51	12.285	3	0.75	H	**15**
		3115 to 4052m	1464.9	23.296	4	1.00	VH	
		4052 to 4855m	2072.3	32.954	2	0.50	M	
		4855 to 6442m	1978.6	31.465	1	0.25	L	
3.	**LULC (Km^2)**	Water bodies	10.182	0.1571	2	0.50	M	**10**
		Tree	377.83	5.8291	1	0.25	L	
		Flooded Vegetation	0.0949	0.0015	1	0.25	L	
		Agriculture	1.183	0.0183	2	0.50	M	
		Built-up	36.439	0.5622	1	0.25	L	

		Bare ground	2058.3	31.754	4	1.00	VH	
		Snow/Ice	742.56	11.456	1	0.25	L	
		Rangeland	3255.3	50.222	3	0.75	H	
4.	**Aspect**	Flat	2.2006	0.035	1	0.25	L	**5**
		North	400.41	6.3675	2	0.5	M	
		Northeast	830.01	13.199	3	0.75	H	
		East	761.25	12.106	3	0.75	H	
		Southeast	819.94	13.039	4	1.00	VH	
		South	754.85	12.004	4	1.00	VH	
		Southwest	842.09	13.391	1	0.25	L	
		West	696.76	11.08	3	0.75	H	
		Northwest	805.66	12.812	3	0.75	H	
		North	375.18	5.9663	2	0.5	M	
5.	**Geology**	Y ; Granite & Granitoid	1340.4	20.679	1	0.25	L	**10**
		Pt2 ; Ortho quartzite, basic volcanic	248.14	3.828	3	0.75	H	
		CJ ; Limestone, dolomite, Shale, Sandstone,	420.89	6.493	4	1.00	VH	
		OC ; Limestone, Siltstone, shale,	1122.6	17.319	2	0.5	M	
		Pt3e ; Greenish grey sandstone,	2012.1	31.041	1	0.25	L	
		Pt1 ; Regionally metamorphosed	1264.2	19.502	1	0.25	L	
		Fault	73.647	1.1361	4	1.00	VH	
6.	**Soil Texture**	Others	4141.4	63.961	1	0.25	L	**5**
		Coarse Texture	1215.6	18.774	4	1.00	VH	
		Fine texture	669.92	10.347	2	0.50	M	
		Medium texture	447.93	6.918	3	0.75	H	
7.	**Precipitation (mm)**	196-448	671.81	10.46	1	0.25	L	**15**
		448-662	1013.7	15.79	4	1.00	VH	
		662-1016	403.21	6.28	2	0.50	M	
		1016-1642	4328.5	67.45	3	0.75	H	
8.	**Stream Order**	First	54.272343	54.365359	1	0.25	L	**5**
		Second	22.709218	22.748139	1	0.25	L	
		Third	12.283906	12.304959	2	0.50	M	
		Fourth	5.6715625	5.6812828	3	0.75	H	
		Fifth	2.55125	2.5556225	3	0.75	H	
		Sixth	1.8475	1.8506663	4	1.00	VH	
		Seventh	0.493125	0.4939701	4	1.00	VH	

The eight reclassified raster layers were subsequently integrated using the GIS Weighted Overlay tool. Higher combined evaluations identified cells in which the selected conditioning factors collectively represented greater relative susceptibility. The resulting district-scale surface was grouped into four interpretive classes — Low, Medium, High and Very High, and

is presented in Figure 17. Table 1 provides the full record of factor subclasses, ranks and percentage influences used to produce this surface. No independent statistical validation, such as ROC/AUC or a training–validation split, was undertaken; consequently, the output is interpreted throughout the dissertation as relative susceptibility rather than a probability of future landslide occurrence.

A separate spatial interpolation was undertaken to examine the geographical pattern of mapped landslide size. Landslide point features associated with mapped landslide-area values were used as the interpolation input, and Kriging was applied to generate a continuous surface from those observed values. Figure 18 therefore represents variation in the mapped size of already identified landslides rather than variation in the environmental conditioning factors used in the susceptibility model. For this reason, the interpolation is treated as an exploratory landslide-area surface and not as a second susceptibility assessment or a validation of Figure 17. The detailed Kriging variant, semivariogram model and search-neighbourhood settings were not retained in the available methodological record, so these parameters are not reconstructed or inferred here.

4. Results

4.1 Landslide Inventory and Road-Proximity Pattern

A total of 179 mapped landslides were identified within the 1000 m analytical corridor between Kafaur and Khab. Their distribution showed a clear decrease with increasing distance from NH-5. Of the total inventory, 102 landslides (57.0%) occurred within the 0–300 m band, compared with 44 (24.6%) within 300–600 m and 33 (18.4%) within 600–1000 m (Figure 14). The immediate 0–300 m band therefore contained more than half of all mapped corridor landslides. The corresponding raster-based landslide areas also declined outwards, from 3.2884 km² in the first band to 1.8794 km² and 1.1348 km² in the second and third bands respectively. Together, these raster-derived areas totalled 6.3026 km² (Figure 11).

The polygon inventory also showed considerable variation in mapped landslide size. It contained 17 large landslides ranging from 0.101973 to 0.368718 km², 104 medium

landslides ranging from 0.010033 to 0.087193 km², and 58 small landslides ranging from 0.000837 to 0.009940 km². Medium-sized features therefore formed the largest group within the inventory. Prominent mapped locations along the corridor included Shudarang, Powari, Pangi, Tapri and Urni. Figure 13 presents author photographs taken along NH-5 on 15 April 2025 as supplementary visual context for selected landslide locations; the photographs were not used as analytical or validation data.

At the wider district scale, the existing GIS dataset contained 1,394 mapped landslide occurrences. Of these, 179 were located within the 1000 m NH-5 analytical corridor and 1,215 outside it (Figure 12). As the source and mapping history of the wider inventory are less clearly documented than the corridor inventory, these figures are retained only as district-scale spatial context and are not treated as directly equivalent to the corridor mapping.

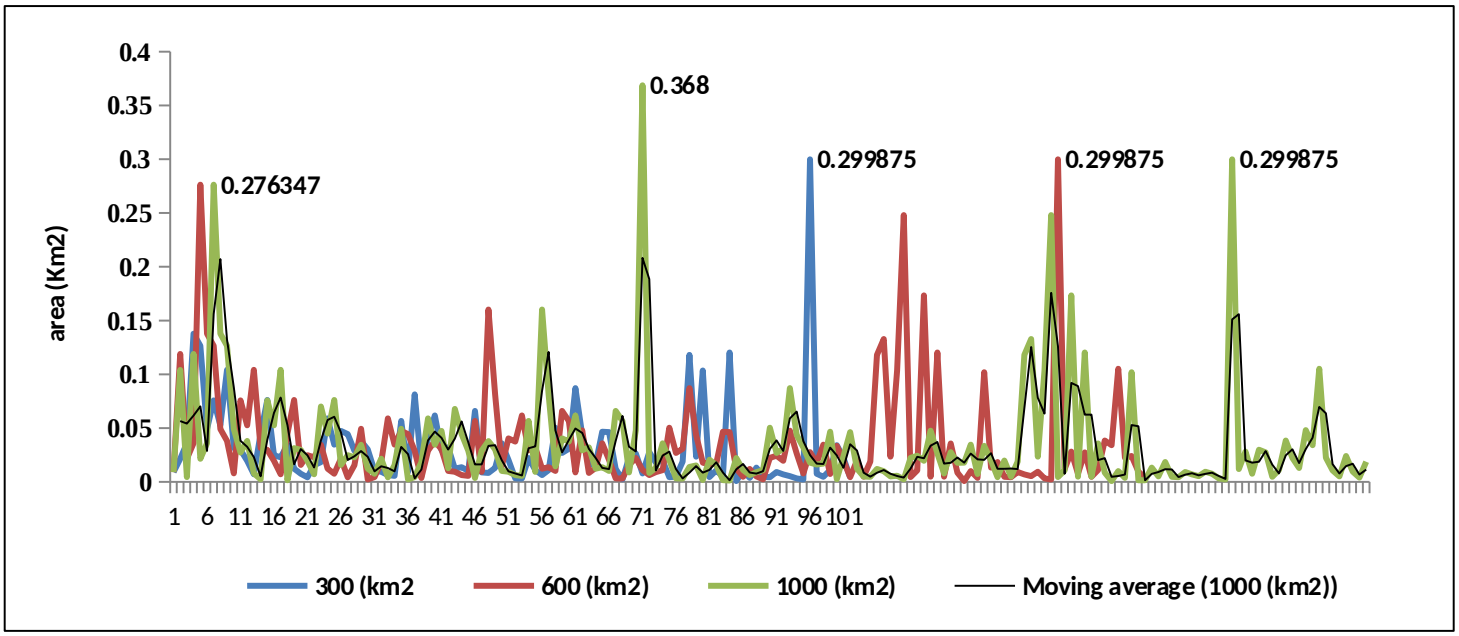


Figure 11 Distribution of mapped landslide-area values across NH-5 buffer distances

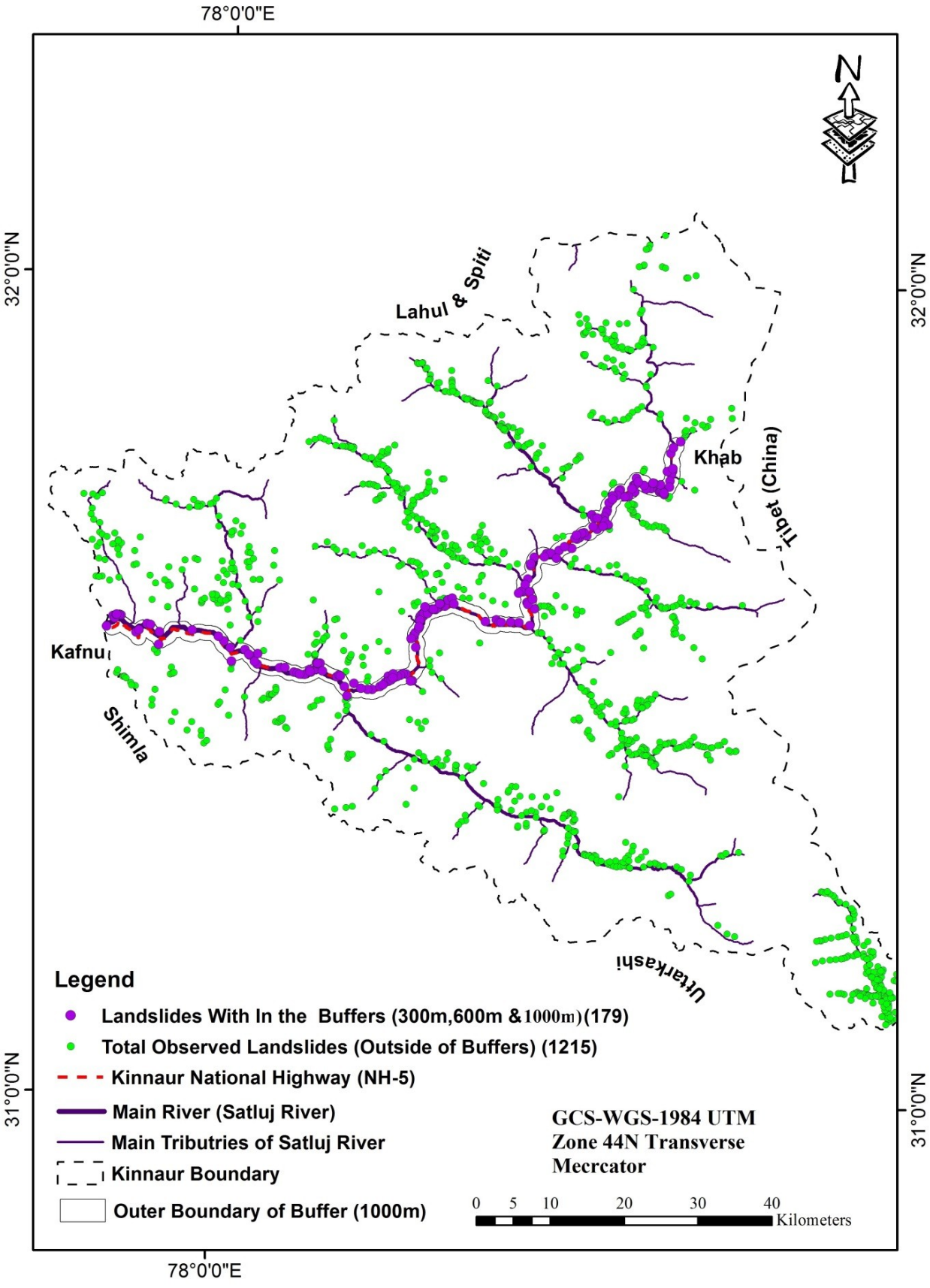


Figure 12 Mapped landslides within and outside the NH-5 analytical corridor



Figure 13 Selected author photographs of landslides along NH-5, taken during a personal visit in April 2025 for contextual illustration.

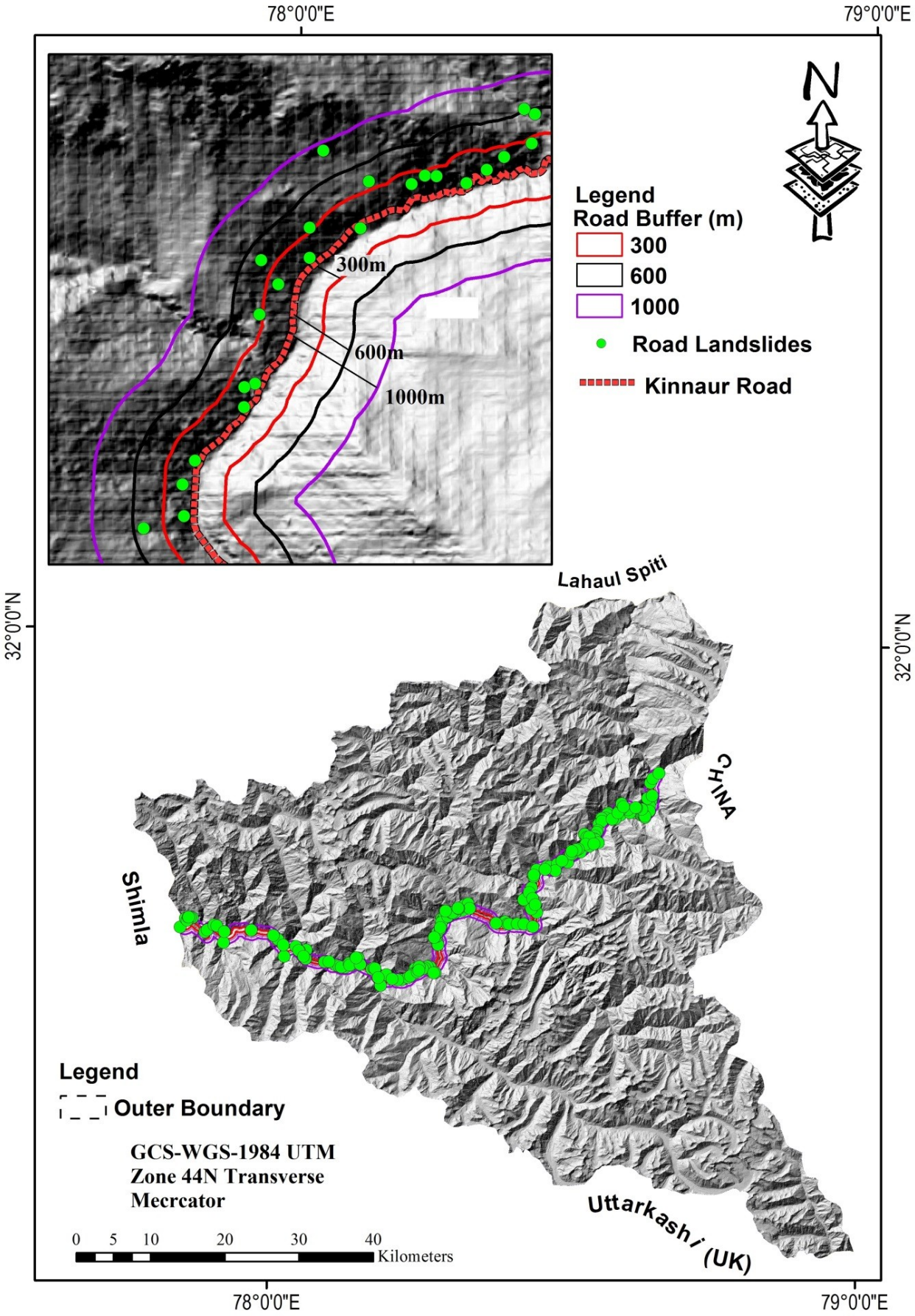


Figure 14 Mapped landslides within the 1000 m NH-5 corridor

4.2 Land-Cover and Settlement Exposure

Within the 1000 m NH-5 analytical corridor, mapped landslides overlapped a total of 6.3026 km² of the classified land-cover surface (Table 2). Rangeland accounted for the largest share of this overlap at 3.6614 km² (58.09%), followed by bare ground at 2.4016 km² (38.10%). Together, these two classes represented approximately 96.2% of the total corridor overlap. Built-up land accounted for a further 0.1943 km² (3.08%), while considerably smaller overlaps were recorded for water bodies (0.0352 km²), tree cover (0.0072 km²) and agriculture (0.0029 km²).

The amount of mapped LULC overlap also decreased across the three road-distance bands, from 3.2884 km² within 0–300 m to 1.8794 km² within 300–600 m and 1.1348 km² within 600–1000 m. Rangeland formed the largest component in each band, decreasing from 1.8836 km² in the innermost band to 1.1281 km² and 0.6497 km² in the two outer bands. Bare ground showed a similar pattern, with respective overlaps of 1.2712 km², 0.6863 km² and 0.4441 km². Built-up overlap also declined from 0.0970 km² within 0–300 m to 0.0645 km² and 0.0328 km² further from the highway. The distribution of the individual land-cover classes is shown in Figure 15.

Table 2 LULC overlap with mapped landslides along the NH-5 corridor

S.No.	Types	0-300m	300-600m	600-1000m	Total overlap (km²)	Share of overlap (%)
1	Water bodies	0.0347	0.0005	0	0.0352	0.5585
2	Tree	0.0019	0	0.0053	0.0072	0.114239
3	Agriculture	0	0	0.0029	0.0029	0.046013
4	Built-up	0.097	0.0645	0.0328	0.1943	3.082855
5	Bare ground	1.2712	0.6863	0.4441	2.4016	38.10491
6	Rangeland	1.8836	1.1281	0.6497	3.6614	58.09349
		3.2884	**1.8794**	**1.1348**	**6.3026**	**100**

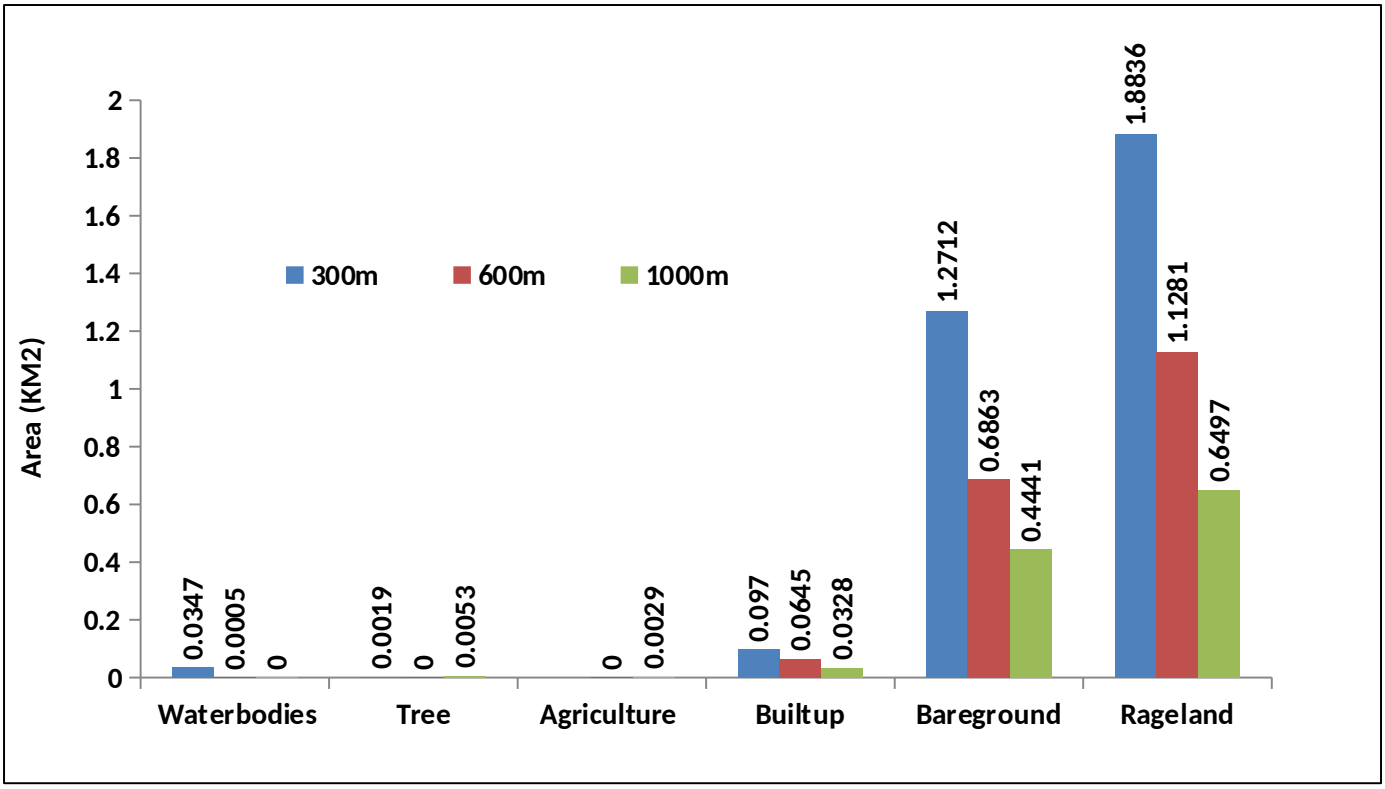


Figure 15 Land-cover overlap within the NH-5 corridor

At the wider Kinnaur scale, mapped landslide–LULC overlap totalled 60.0829 km² (Table 3). Rangeland covered 50.22% of the wider study area and accounted for 37.77 km² (62.87%) of the mapped overlap, while bare ground covered 31.75% of the study area and contributed 19.89 km² (33.11%) of the overlap. Smaller mapped overlaps were recorded within snow/ice (1.63 km²), tree cover (0.384 km²), built-up land (0.29 km²), water bodies (0.102 km²) and agriculture (0.0032 km²), with no recorded overlap in flooded vegetation.

Table 3 District-scale LULC overlap with mapped landslides

S.No.	Types	Total Area (km^2)	% (Area)	Mapped landslide overlap area (km^2)	Mapped overlap % (Area)
1	Water bodies	10.1818	0.157	0.102	0.17
2	Tree	377.83	5.82	0.384	0.63
3	Flooded Vegetation	0.0949	0.0014	0	
4	Agriculture	1.183	0.018	0.0032	0.0053
5	Built-up	36.43	0.562	0.29	0.485
6	Bare ground	2058.26	31.75	19.89	33.106
7	Snow/Ice	742.55	11.45	1.63	2.715
8	Rangeland	3255.28	50.22	37.77	62.87
	Total	**6481.8405**	**100**	**60.0829**	**100**

Settlement mapping identified 91 village point locations within the 1000 m analytical corridor. These were mapped in relation to NH-5 and the landslide inventory, including settlements such as Pangi, Powari, Tapri and Urni (Figure 16). Their inclusion identifies locations with potential spatial exposure within the corridor; it does not establish that all 91 settlements or individual buildings experienced direct landslide damage.

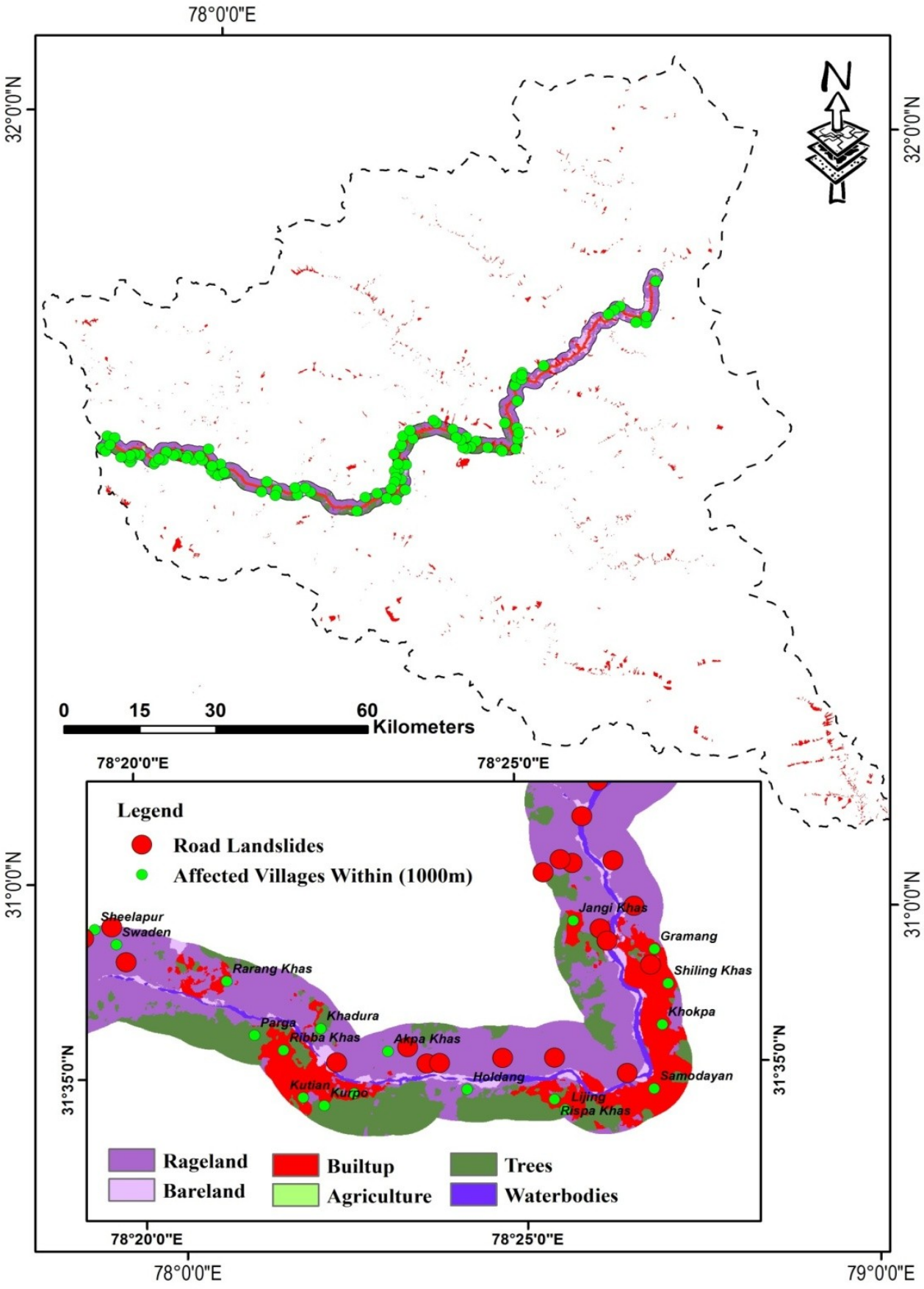


Figure 16 Settlement locations within the 1000 m NH-5 corridor

4.3 Relative Susceptibility and Spatial Pattern of Landslide Area

The relative susceptibility assessment integrated eight reclassified conditioning factors across the wider Kinnaur study region. Within these factor classifications, the highest relative rank was assigned to slopes of 45.04–85.07°, the 3115–4052 m elevation band, bare ground, southeast and south-facing slopes, geological faults and the CJ geological unit, coarse soil texture, the 448–662 mm precipitation class, and sixth- and seventh-order streams. These ranked subclasses, together with the remaining lower-ranked classes, formed the inputs to the final Weighted Overlay assessment.

The Weighted Overlay classified the study region into four relative landslide-susceptibility classes (Figure 17; Table 4). Medium susceptibility occupied the largest area, covering 2,890.66 km² (44.60%). This was followed by Low susceptibility at 1,729.00 km² (26.67%) and High susceptibility at 1,690.93 km² (26.09%). Very High susceptibility occupied the smallest area at 171.25 km² (2.64%). High and Very High susceptibility together accounted for approximately 28.73% of the analytical area, while all four classes together represented the complete 6,481.8405 km² study region.

Table 4 Relative susceptibility class areas in Kinnaur

S. No	Susceptibility	Area (Km^2)	Area (%)
1.	Very High	171.251875	2.642025
2.	High	1690.931875	26.08722
3.	Medium	2890.660156	44.59629
4.	Low	1728.996594	26.67447
		6481.8405	**100**

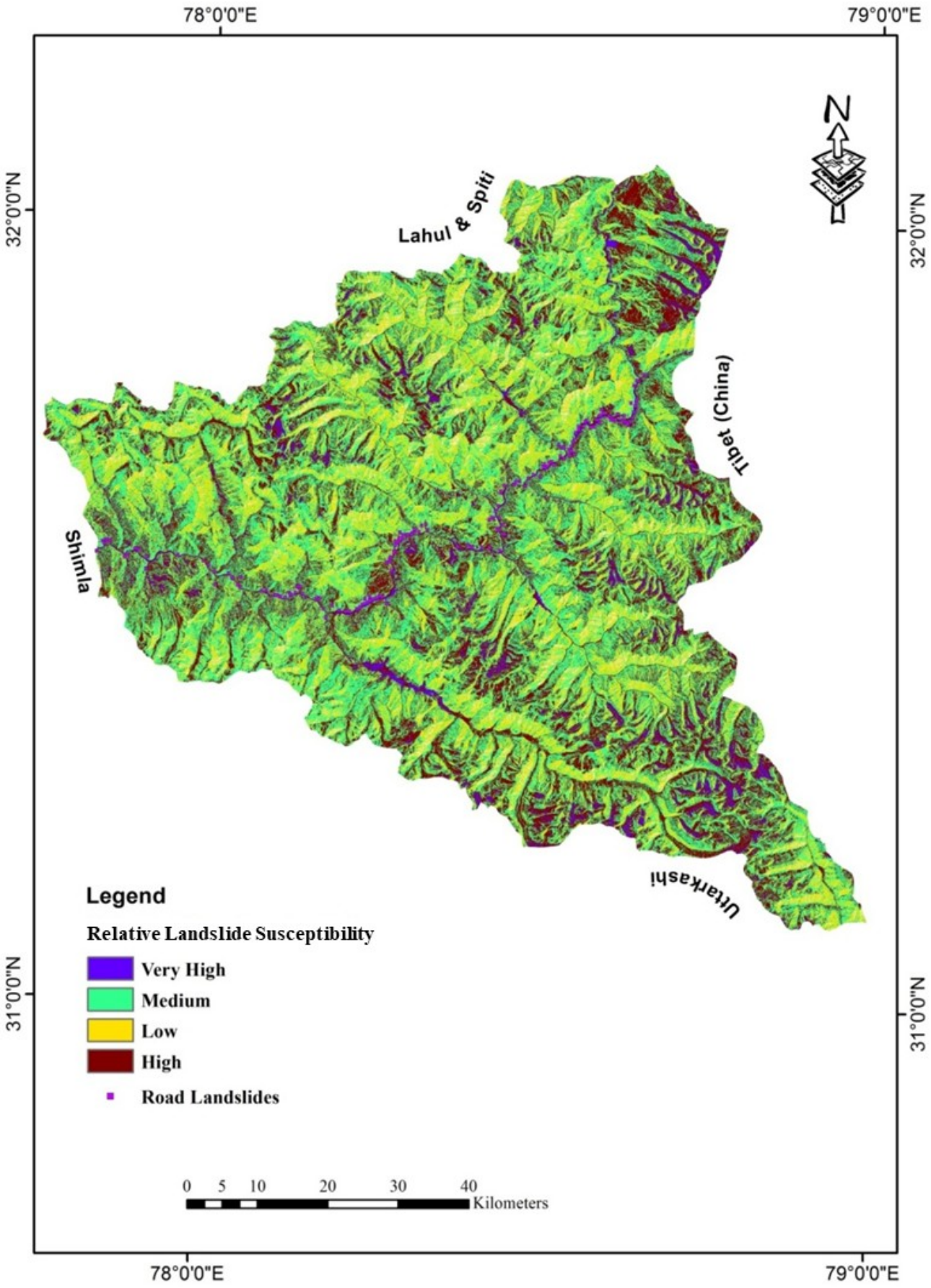


Figure 17 Relative landslide susceptibility across Kinnaur

A separate Kriging interpolation of mapped landslide-area values was used to examine the spatial variation in mapped landslide size (Figure 18). The interpolated surface was classified as Low (0.0011–0.0915 km²), Medium (0.0915–0.216 km²), High (0.216–0.532 km²) and Very High (0.532–1.443 km²). Higher interpolated values occurred as localised patches rather than forming a continuous pattern across the study region. Unlike Figure 17, this surface is based on the areas of already mapped landslides rather than the eight environmental conditioning factors. It therefore represents an exploratory spatial pattern of landslide area and is not treated as a second susceptibility model or as validation of the Weighted Overlay result.

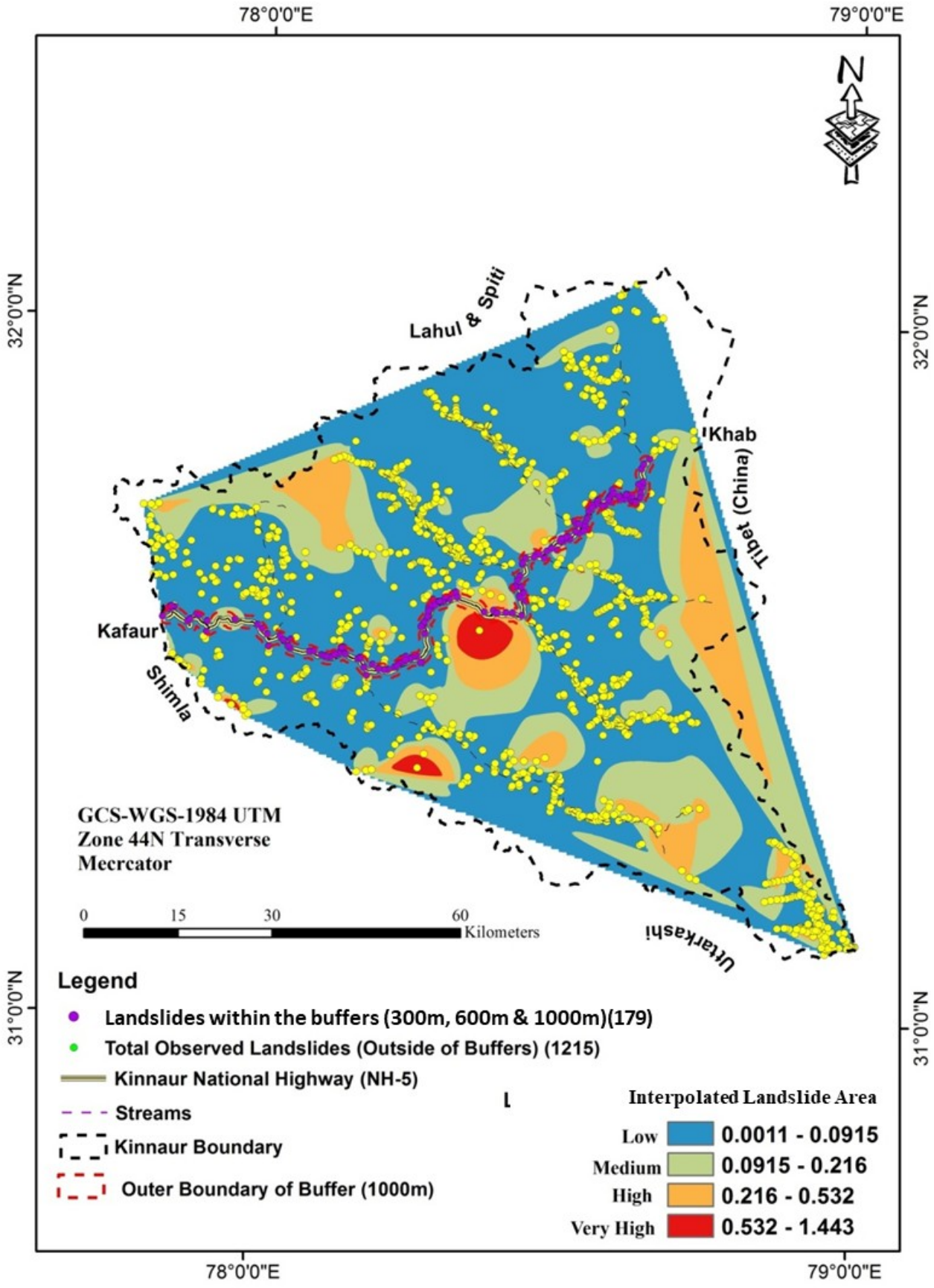


Figure 18 Interpolated spatial pattern of mapped landslide area

5. Discussion

5.1 Road-Proximity and Landslide Distribution

The corridor analysis shows a clear decrease in mapped landslide occurrence with increasing distance from NH-5. Of the 179 landslides identified within the 1000 m analytical corridor, 102 (57.0%) occurred within 0–300 m, compared with 44 (24.6%) between 300–600 m and 33 (18.4%) between 600–1000 m. The mapped landslide area followed the same outward pattern, declining from 3.2884 km² in the innermost band to 1.8794 km² and 1.1348 km² in the two outer bands. More than half of the mapped corridor inventory was therefore concentrated within 300 m of the highway. This demonstrates a strong spatial association between mapped landslides and the immediate NH-5 corridor, although the analysis does not by itself establish road construction as the cause of these failures. This caution is consistent with the multi-factor view of slope failure described by Varnes (1978), in which instability can result from the combined influence of processes that increase shear stress and conditions that reduce the shear strength of slope materials.

The observed pattern fits the wider landslide history of the Satluj Valley. Gupta and Sah (2008a) documented a substantial increase in active landslide coverage along the valley between the 1990s and 2006 and related this change to a combination of naturally susceptible terrain and increasing human disturbance. Their study identified road excavation, widening and blasting as potential contributors to slope instability, while also recognising the importance of lithology, river erosion, rainfall and other geomorphological conditions. They reported important landslide locations around Tapri, Shongtong, Powari and Pangi, several of which also occur within or close to the present inventory. This suggests that the road-proximal pattern observed here occurs within a corridor with a longer documented history of recurring slope instability rather than representing an isolated recent condition.

Regional work provides similar context. Jamwal et al. (2020) showed that landslide vulnerability in Kinnaur reflects the combined influence of steep terrain, lithology, drainage, land cover and anthropogenic activity. This is particularly relevant along the deeply incised Satluj Valley, where road infrastructure and settlements occupy a comparatively narrow

mountainous corridor. The clustering observed close to NH-5 can therefore be understood as occurring where transport infrastructure is positioned within an already sensitive geomorphological setting, rather than as evidence of a single controlling factor.

Urni illustrates this interaction at a more local scale. Kumar et al. (2019) described the Urni landslide as a complex failure developed on the steep Satluj valley side, where movement partially dammed the river and damaged approximately 200 m of NH-5. Its development involved geological, topographic and rainfall controls, reinforcing the need to interpret highway proximity together with natural slope conditions. A comparable Himalayan road-corridor study by Sur et al. (2020) also identified greater susceptibility close to roads, although their fuzzy-AHP approach differs from the inventory-based distance analysis used in this dissertation.

The road-proximity pattern is therefore useful for identifying sections of the corridor that may require greater attention, but the result remains descriptive. The three bands report mapped counts and areas rather than an area-normalised landslide density or statistically tested road effect. In addition, the 1000 m corridor is an analytical window rather than a physical limit to landslide influence. The pattern is consequently best interpreted as the spatial interaction of NH-5 with steep, geologically complex and environmentally sensitive valley-side terrain.

5.2 Environmental Susceptibility and Landslide-Area Pattern

The relative susceptibility assessment indicates considerable spatial variation across Kinnaur. Medium susceptibility formed the largest class (44.60%), while High and Very High susceptibility together occupied 28.73% of the analytical area. Slope received the greatest influence in the model (35%), reflecting its importance as a major terrain-conditioning factor. Steeper slopes generally increase the potential for gravitational instability, although slope angle alone does not determine failure (Çellek, 2022). This emphasis is consistent with Himalayan susceptibility studies in which slope has been treated as an important conditioning variable, including the road-corridor assessment of Sur et al. (2020) and the Satluj-based analysis of Kuniyal et al. (2019). The present model assigned the highest rank to

slopes of 45.04–85.07°, while Kuniyal et al. (2019) also reported high landslide occurrence on very steep slopes in the Satluj basin.

Elevation showed a more complex relationship. The 3115–4052 m band received the highest rank in the present model, but regional studies have also reported substantial landslide activity at lower and middle elevations where roads, settlements and development are concentrated (Jamwal et al., 2020). This difference is important because elevation itself is not a direct trigger; instead, it influences other environmental conditions such as vegetation, weathering and precipitation. The elevation ranking should therefore be interpreted as part of the combined model rather than as an independent indicator of failure.

The LULC and aspect classifications show closer agreement with previous regional work. Bare ground received the highest susceptibility rank and rangeland a high rank, while tree cover was ranked lower. Jamwal et al. (2020) similarly reported considerable landslide-related degradation on barren and grassland surfaces. South and southeast aspects also received the highest ranks in the present model, broadly corresponding with patterns reported by Kuniyal et al. (2019). These relationships are likely to reflect the combined effects of terrain orientation, vegetation condition, moisture availability and weathering rather than aspect or land cover acting alone.

Geological structure and material properties add further uncertainty. Faults and selected lithological units received high rankings, consistent with evidence that structural discontinuities contribute to instability in Kinnaur (Joshi et al., 2024). However, the geological classification also demonstrates a limitation of the knowledge-driven ranking approach. Greenish-grey sandstone (Pt3e), for example, was ranked Low in the present model, whereas Jamwal et al. (2020) reported substantial landslide and degradation occurrence within this unit. This difference suggests that the assigned subclass ranks do not reproduce every locally observed geological pattern and should therefore be interpreted cautiously. Soil texture was included because material characteristics can influence erosion and slope response (Ravishankar and Sreenivas, 2010), although the 1:50,000 NBSS dataset cannot represent site-specific properties such as local cohesion or discontinuity conditions.

Precipitation and drainage also require careful interpretation. Rainfall is an important trigger of Himalayan landslides (Dikshit et al., 2020), but the 0.25° PERSIANN-CDR layer used here represents broad annual rainfall conditions rather than individual short-duration, high-intensity events (Ashouri et al., 2015). This distinction is relevant in Himachal Pradesh, where Manocha et al. (2026) emphasise that landslide occurrence may remain high even where broader rainfall totals alone do not explain the observed temporal pattern. Stream order similarly represents drainage hierarchy and the broader valley structure, but does not directly model discharge, river erosion or flood behaviour. Gupta and Sah (2008b) demonstrated the importance of river erosion and toe-cutting within the Satluj system, providing useful context for including drainage-related conditions without implying that the present static stream-order layer models these processes directly.

At the integrated model level, the 28.73% combined High and Very High susceptibility area is broadly comparable with the regional pattern reported by Jamwal and Sharma (2022). However, this similarity should not be considered independent validation because both assessments use knowledge-driven spatial weighting. Statistical approaches applied by Qazi et al. (2023), Lee and Pradhan (2007) and Chen et al. (2015) derive factor relationships more directly from observed landslide distributions and therefore provide useful alternatives for future testing of the present weighting scheme. Finally, Figure 18 remains separate from the Weighted Overlay result. It represents an exploratory Kriging interpolation of mapped landslide area and neither validates Figure 17 nor represents landslide runout, which would require a different modelling approach (Mergili et al., 2019).

5.3 Exposure, Management Implications and Uncertainty

The exposure analysis shows that mapped landslides overlap mainly with rangeland and bare ground within the NH-5 corridor. Rangeland represented 58.09% of the 6.3026 km² mapped LULC overlap, while bare ground accounted for 38.10%. Built-up land represented a much smaller share (3.08%), and agricultural overlap was limited to 0.0029 km². A similar pattern occurs at the wider Kinnaur scale, where rangeland and bare ground also form most of the mapped landslide–LULC overlap. This broadly agrees with regional work showing that sparsely vegetated and barren surfaces form a substantial part of Kinnaur and occur widely within landslide-affected terrain (Jamwal et al., 2020). However, these classes also occupy

large proportions of the district, so their dominance in the overlap should not be interpreted as evidence that they have the highest landslide rate without area-normalised analysis. Similarly, the relatively small mapped overlaps of built-up and agricultural land do not necessarily imply that their consequences are unimportant, because the present analysis measures spatial extent rather than the value or vulnerability of the exposed assets.

The settlement results add an important human dimension to this pattern. A total of 91 village point locations were identified within the 1000 m analytical corridor and mapped in relation to NH-5 and the landslide inventory. Their occurrence within the same narrow valley corridor indicates that settlements, transport connectivity and unstable slopes are spatially concentrated in close proximity. Regional work has also highlighted the concentration of settlements and infrastructure within the deeply incised Satluj Valley (Jamwal et al., 2020). Road-corridor studies elsewhere in the Himalaya have shown the value of relating susceptibility information to nearby settlements for spatial prioritisation (Sur et al., 2020), while Martha et al. (2021) demonstrated how landslide inventories can be combined with population and infrastructure information when assessing exposure at broader scales. In the present study, however, settlements are represented as geographic points rather than village boundaries or building footprints. The 91 locations therefore indicate potential spatial exposure and proximity, not confirmation that every village, household or structure experienced direct landslide damage.

These patterns have practical implications for corridor management. Combining road-proximity, LULC exposure and relative susceptibility information provides a useful screening framework for identifying sections of NH-5 where more detailed field checks and slope investigations may be required. Similar GIS-based assessments in the Satluj basin have been used to support spatial planning and the identification of environmentally sensitive areas (Kuniyal et al., 2019; Jamwal and Sharma, 2022). In areas dominated by bare ground and sparsely vegetated rangeland, vegetation restoration, afforestation and other bioengineering measures could be considered as part of wider slope-management strategies, particularly where surface erosion is evident (Jamwal and Sharma, 2022). Small-scale drainage and erosion-control measures, including check dams where appropriate, may also be considered following site-specific field and engineering assessment. Along road-cut sections, however, the GIS outputs should be used to identify areas for further investigation

rather than to prescribe particular engineering structures, because local rock structure, drainage and slope geometry can vary substantially over short distances (Singh et al., 2022).

Several uncertainties limit the interpretation of the results. Manual interpretation of November 2025 imagery may omit small, old, shaded or vegetation-covered failures, since optical inventory mapping depends on image quality and feature visibility (Kumar et al., 2019). The 0.25° PERSIANN-CDR precipitation layer provides broad regional rainfall information but is coarse relative to the strong local topographic variation of Kinnaur (Ashouri et al., 2015). The knowledge-driven Weighted Overlay was also not independently validated, so its classes represent relative susceptibility rather than predictive probabilities (Dikshit et al., 2020). In addition, the 1000 m corridor is an analytical boundary rather than a maximum physical landslide-impact distance. The study does not model downslope runout or cascading impacts, which would require a different modelling approach (Mergili et al., 2019). Figure 18 is likewise exploratory and should not be interpreted as validation of the susceptibility map.

Future work could therefore strengthen the analysis through systematic field validation, higher-resolution environmental and settlement datasets, and temporal deformation monitoring. PS-InSAR, which has already been applied in Kinnaur, provides one example of how repeated observations could complement a static optical inventory by identifying active ground movement through time (Joshi et al., 2024).

6. Conclusion

This dissertation assessed the spatial distribution of mapped landslides, relative susceptibility and associated exposure along the NH-5–Satluj River corridor from Kafaur to Khab, while using the wider Kinnaur District to provide regional susceptibility context. The GIS and remote-sensing approach combined corridor-based landslide inventory mapping, road-distance analysis, LULC and settlement exposure, and a district-scale Weighted Overlay assessment.

A total of 179 mapped landslides were identified within the 1000 m analytical corridor. Their distribution showed a clear decline with increasing distance from NH-5, with 102 landslides (57.0%) within 0–300 m, compared with 44 (24.6%) within 300–600 m and 33 (18.4%) within

600–1000 m. This indicates a strong road-proximal spatial pattern, although it does not establish that road construction caused the mapped failures or is the sole reason for it.

Across wider Kinnaur, Medium susceptibility formed the largest class (44.60%), while High and Very High susceptibility together occupied 28.73% of the analytical area. These patterns reflect the combined influence of the eight selected conditioning factors, with slope receiving the greatest model influence. However, the output represents relative, expert-informed susceptibility rather than an independently validated prediction, while the Kriging surface represents only the spatial pattern of mapped landslide area.

The exposure analysis showed that rangeland and bare ground dominated the 6.3026 km^2 of mapped LULC overlap, while 91 village point locations occurred within the analytical corridor. These results represent spatial exposure rather than confirmed damage or quantified risk. Overall, combining corridor-focused inventory and exposure analysis with wider susceptibility mapping provides a useful basis for prioritising field inspection and more detailed slope assessment. Where supported by site-specific investigation, vegetation restoration, bioengineering and small-scale drainage measures may also be considered. Future work could strengthen the study through higher-resolution environmental data, systematic field validation and temporal deformation monitoring such as PS-InSAR.

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8. Appendices

1. Large Landslides area wise (0.368718 to 0.101973 km²)

S.No.	Landslides (Large km²)	Latitude	Longitude
1	0.368718	31° 40' 37.476" N	78° 29' 46.856" E
2	0.299875	31° 35' 7.080" N	78° 22' 32.792" E
3	0.276347	31° 35' 6.777" N	78° 24' 43.524" E
4	0.247978	31° 41' 35.506" N	78° 30' 9.445" E
5	0.173384	31° 33' 57.309" N	77° 57' 29.408" E
6	0.160197	31° 44' 50.829" N	78° 35' 3.648" E
7	0.137837	31° 35' 5.868" N	78° 25' 24.523" E
8	0.132846	31° 41' 21.990" N	78° 30' 2.918" E
9	0.126479	31° 34' 53.709" N	78° 26' 21.354" E
10	0.120235	31° 33' 52.096" N	77° 52' 57.975" E
11	0.118935	31° 35' 16.393" N	78° 23' 29.009" E
12	0.118057	31° 42' 14.364" N	78° 31' 34.632" E
13	0.105225	31° 47' 44.198" N	78° 38' 4.802" E
14	0.104119	31° 40' 12.258" N	78° 27' 38.635" E
15	0.104022	31° 43' 6.266" N	78° 31' 25.164" E
16	0.103522	31° 41' 43.527" N	78° 30' 36.578" E
17	0.101973	31° 33' 45.472" N	77° 54' 43.493" E

2. Medium Landslides area wise (0.087193 to 0.010033 km²)

S.No.	Area (km²)	Latitude	Longitude
1	0.087193	31° 31' 1.684" N	78° 15' 23.377" E
2	0.081391	31° 45' 0.026" N	78° 33' 49.946" E
3	0.076175	31° 31' 8.131" N	78° 15' 53.805" E
4	0.075762	31° 38' 17.903" N	78° 26' 5.429" E
5	0.070082	31° 30' 56.484" N	78° 14' 21.844" E
6	0.067884	31° 36' 45.840" N	78° 19' 38.935" E
7	0.06606	31° 42' 38.490" N	78° 32' 34.245" E
8	0.061825	31° 44' 1.315" N	78° 32' 52.701" E
9	0.059012	31° 36' 3.453" N	78° 18' 28.831" E
10	0.056717	31° 45' 15.794" N	78° 36' 0.464" E
11	0.05651	31° 40' 12.891" N	78° 27' 59.178" E
12	0.052585	31° 39' 53.330" N	78° 26' 35.423" E
13	0.050441	31° 31' 23.057" N	78° 5' 29.958" E
14	0.049395	31° 35' 37.072" N	78° 17' 16.192" E
15	0.04905	31° 36' 34.907" N	78° 26' 3.686" E
16	0.048105	31° 46' 23.504" N	78° 37' 49.997" E
17	0.047855	31° 41' 32.746" N	78° 31' 2.608" E
18	0.047606	31° 45' 4.099" N	78° 36' 2.459" E
19	0.047327	31° 36' 10.502" N	78° 19' 3.760" E
20	0.04668	31° 30' 40.612" N	78° 14' 36.537" E

21	0.046204	31° 29' 56.551" N	78° 12' 37.064" E
22	0.044971	31° 30' 51.535" N	78° 14' 52.097" E
23	0.044476	31° 36' 21.263" N	78° 19' 49.403" E
24	0.043613	31° 30' 57.024" N	78° 15' 8.124" E
25	0.04046	31° 44' 33.705" N	78° 33' 34.217" E
26	0.038411	31° 45' 24.599" N	78° 37' 15.168" E
27	0.038106	31° 37' 21.233" N	78° 25' 43.247" E
28	0.037954	31° 46' 10.581" N	78° 38' 10.875" E
29	0.037493	31° 44' 31.688" N	78° 33' 9.867" E
30	0.036011	31° 35' 4.555" N	78° 23' 53.879" E
31	0.035818	31° 42' 53.484" N	78° 31' 54.453" E
32	0.035754	31° 34' 19.571" N	77° 53' 37.402" E
33	0.034587	31° 44' 53.254" N	78° 37' 30.478" E
34	0.034574	31° 36' 8.534" N	78° 18' 39.133" E
35	0.034502	31° 34' 1.256" N	78° 16' 28.745" E
36	0.034093	31° 46' 37.407" N	78° 38' 18.195" E
37	0.033561	31° 39' 51.841" N	78° 28' 30.157" E
38	0.032135	31° 43' 27.596" N	78° 32' 45.693" E
39	0.032037	31° 31' 7.233" N	78° 14' 46.358" E
40	0.031369	31° 30' 55.507" N	78° 7' 12.301" E
41	0.030809	31° 30' 34.504" N	78° 14' 22.208" E
42	0.030226	31° 45' 42.411" N	78° 38' 3.746" E
43	0.029943	31° 42' 23.266" N	78° 32' 51.879" E
44	0.029708	31° 31' 6.114" N	78° 7' 44.673" E
45	0.029359	31° 44' 17.070" N	78° 33' 10.948" E
46	0.028958	31° 46' 25.538" N	78° 38' 17.001" E
47	0.02888	31° 30' 55.913" N	78° 10' 8.160" E
48	0.028179	31° 44' 28.686" N	78° 34' 14.029" E
49	0.027711	31° 45' 24.788" N	78° 35' 9.386" E
50	0.027332	31° 42' 16.653" N	78° 32' 25.001" E
51	0.027304	31° 36' 8.696" N	78° 26' 42.344" E
52	0.026902	31° 30' 30.189" N	78° 14' 7.976" E
53	0.026388	31° 45' 21.346" N	78° 35' 51.066" E
54	0.024839	31° 33' 7.356" N	78° 16' 38.466" E
55	0.024357	31° 44' 53.538" N	78° 37' 11.718" E
56	0.024102	31° 36' 49.986" N	78° 26' 31.073" E
57	0.023802	31° 30' 21.678" N	78° 11' 26.934" E
58	0.023491	31° 41' 47.040" N	78° 30' 48.734" E
59	0.022896	31° 33' 49.434" N	78° 16' 29.259" E
60	0.022429	31° 48' 8.097" N	78° 38' 18.113" E
61	0.022395	31° 45' 15.567" N	78° 38' 2.504" E
62	0.022246	31° 45' 30.788" N	78° 36' 42.575" E
63	0.022144	31° 31' 4.493" N	78° 6' 16.655" E
64	0.021524	31° 35' 24.927" N	78° 16' 58.138" E
65	0.021481	31° 35' 4.549" N	78° 23' 43.868" E

66	0.020619	31° 34' 55.252" N	77° 51' 12.479" E
67	0.020436	31° 37' 53.539" N	78° 25' 52.118" E
68	0.019875	31° 37' 22.067" N	78° 26' 15.576" E
69	0.019873	31° 39' 57.038" N	78° 27' 27.075" E
70	0.019601	31° 44' 56.493" N	78° 37' 16.079" E
71	0.019585	31° 31' 17.259" N	78° 8' 6.345" E
72	0.018798	31° 41' 18.433" N	78° 30' 12.600" E
73	0.018712	31° 36' 5.814" N	78° 17' 21.328" E
74	0.018514	31° 30' 49.234" N	78° 14' 45.611" E
75	0.018319	31° 35' 53.391" N	78° 17' 54.250" E
76	0.017906	31° 44' 58.493" N	78° 37' 22.040" E
77	0.017896	31° 29' 57.164" N	78° 13' 0.543" E
78	0.017757	31° 44' 42.293" N	78° 37' 35.971" E
79	0.01703	31° 30' 18.481" N	78° 13' 54.993" E
80	0.016093	31° 30' 10.415" N	78° 13' 39.799" E
81	0.015801	31° 31' 9.918" N	78° 15' 37.236" E
82	0.015785	31° 34' 40.107" N	77° 50' 51.930" E
83	0.015534	31° 35' 15.971" N	78° 17' 4.495" E
84	0.015333	31° 31' 39.530" N	78° 8' 26.668" E
85	0.014369	31° 44' 53.291" N	78° 33' 43.048" E
86	0.013934	31° 34' 50.640" N	77° 50' 41.986" E
87	0.013669	31° 42' 58.259" N	78° 32' 31.339" E
88	0.0133	31° 31' 11.757" N	78° 8' 43.544" E
89	0.012968	31° 42' 51.430" N	78° 31' 40.387" E
90	0.01289	31° 45' 32.222" N	78° 36' 36.317" E
91	0.012683	31° 36' 38.682" N	78° 19' 16.527" E
92	0.012297	31° 43' 18.493" N	78° 32' 41.122" E
93	0.012217	31° 34' 10.186" N	78° 16' 33.438" E
94	0.012207	31° 33' 12.758" N	78° 0' 14.639" E
95	0.012074	31° 32' 37.693" N	78° 2' 4.213" E
96	0.011892	31° 44' 30.268" N	78° 34' 32.064" E
97	0.011036	31° 45' 35.715" N	78° 35' 21.623" E
98	0.011034	31° 31' 0.341" N	78° 6' 34.478" E
99	0.01088	31° 31' 8.158" N	78° 7' 39.191" E
100	0.010655	31° 30' 0.745" N	78° 12' 2.333" E
101	0.01063	31° 48' 35.999" N	78° 38' 55.030" E
102	0.010125	31° 45' 8.794" N	78° 36' 15.677" E
103	0.010107	31° 42' 43.676" N	78° 32' 24.595" E
104	0.010033	31° 32' 14.156" N	78° 2' 27.888" E

3. Small Landslides area wise (0.00994 to 0.000837 km²)

S.No.	Area (km²)	Latitude	Longitude
1	0.00994	31° 32' 20.678" N	78° 3' 9.905" E
2	0.009567	31° 39' 25.539" N	78° 26' 18.223" E

3	0.009472	31° 36' 19.490" N	78° 18' 40.371" E
4	0.009446	31° 45' 8.530" N	78° 36' 21.413" E
5	0.009268	31° 45' 19.936" N	78° 34' 57.707" E
6	0.009179	31° 35' 58.668" N	78° 18' 22.303" E
7	0.009014	31° 31' 16.962" N	78° 7' 36.050" E
8	0.009003	31° 41' 24.886" N	78° 30' 25.894" E
9	0.008763	31° 40' 27.327" N	78° 28' 42.783" E
10	0.008488	31° 34' 15.644" N	77° 53' 55.659" E
11	0.008369	31° 42' 3.858" N	78° 31' 55.822" E
12	0.00793	31° 37' 23.925" N	78° 25' 34.027" E
13	0.007794	31° 45' 21.811" N	78° 35' 1.263" E
14	0.007552	31° 36' 21.284" N	78° 18' 36.272" E
15	0.007511	31° 45' 3.812" N	78° 34' 48.735" E
16	0.00751	31° 34' 38.930" N	78° 16' 55.835" E
17	0.007314	31° 44' 48.752" N	78° 37' 34.861" E
18	0.007073	31° 30' 48.859" N	78° 9' 56.838" E
19	0.006968	31° 35' 56.464" N	78° 17' 59.213" E
20	0.00633	31° 45' 7.036" N	78° 36' 28.155" E
21	0.00632	31° 30' 56.204" N	78° 6' 47.658" E
22	0.005732	31° 29' 25.485" N	78° 10' 52.427" E
23	0.005667	31° 45' 6.283" N	78° 37' 2.490" E
24	0.00526	31° 35' 56.325" N	78° 18' 3.589" E
25	0.005183	31° 30' 37.658" N	78° 16' 16.842" E
26	0.005111	31° 31' 46.105" N	78° 8' 17.878" E
27	0.005068	31° 32' 49.597" N	77° 54' 39.628" E
28	0.004976	31° 31' 44.345" N	78° 0' 55.582" E
29	0.004884	31° 39' 17.297" N	78° 26' 18.046" E
30	0.004793	31° 33' 44.866" N	78° 16' 14.180" E
31	0.004786	31° 33' 40.830" N	77° 52' 43.406" E
32	0.004745	31° 32' 44.431" N	78° 0' 47.983" E
33	0.004669	31° 45' 51.593" N	78° 35' 11.194" E
34	0.004667	31° 35' 36.908" N	78° 17' 35.206" E
35	0.004521	31° 35' 54.351" N	78° 17' 38.277" E
36	0.004465	31° 34' 37.588" N	78° 16' 51.792" E
37	0.004461	31° 34' 53.893" N	77° 51' 23.461" E
38	0.004382	31° 36' 26.210" N	78° 26' 8.842" E
39	0.004317	31° 35' 53.722" N	78° 18' 14.748" E
40	0.00414	31° 35' 25.783" N	78° 17' 16.258" E
41	0.004032	31° 40' 44.388" N	78° 29' 10.623" E
42	0.003657	31° 32' 16.859" N	78° 2' 23.575" E
43	0.003642	31° 30' 53.148" N	78° 10' 15.852" E
44	0.003273	31° 31' 46.722" N	78° 2' 54.855" E
45	0.003257	31° 34' 30.102" N	78° 16' 51.746" E
46	0.00309	31° 31' 21.712" N	78° 5' 4.630" E
47	0.002485	31° 30' 38.807" N	78° 16' 25.947" E

48	0.002451	31° 37' 15.328" N	78° 25' 20.189" E
49	0.002335	31° 30' 6.047" N	78° 10' 37.421" E
50	0.002297	31° 33' 32.775" N	77° 59' 46.160" E
51	0.002278	31° 35' 4.668" N	78° 16' 57.424" E
52	0.00178	31° 31' 46.292" N	78° 8' 30.850" E
53	0.001558	31° 31' 47.018" N	78° 8' 13.059" E
54	0.001518	31° 34' 57.006" N	77° 51' 1.708" E
55	0.001461	31° 34' 3.618" N	77° 50' 11.128" E
56	0.001447	31° 34' 55.921" N	77° 50' 58.371" E
57	0.00125	31° 31' 45.085" N	78° 8' 14.305" E
58	0.000837	31° 31' 35.174" N	78° 3' 7.918" E

4. Landslides affected Villages with in the multi-buffer of 300,600 and 1000m

S.No.	Villages	Latitude	Longitude
1	Barang	31° 30' 38.124" N	78° 15' 32.361" E
2	Boktu	31° 34' 36.898" N	78° 16' 36.922" E
3	Kashmir	31° 34' 48.145" N	78° 16' 41.499" E
4	Khwangi	31° 32' 55.117" N	78° 16' 31.788" E
5	Kilba Khas	31° 30' 58.593" N	78° 8' 50.531" E
6	Kothi	31° 32' 47.142" N	78° 16' 9.758" E
7	Pangi	31° 35' 28.474" N	78° 16' 48.355" E
8	Powari	31° 33' 16.318" N	78° 17' 9.586" E
9	Purbani Khas	31° 35' 17.831" N	78° 17' 40.758" E
10	Raang	31° 32' 7.634" N	78° 16' 46.574" E
11	Radule	31° 30' 59.380" N	78° 14' 40.548" E
12	Ragura	31° 35' 55.896" N	78° 17' 24.055" E
13	Rali	31° 29' 36.252" N	78° 12' 57.337" E
14	Reckong Peo	31° 32' 15.499" N	78° 16' 16.370" E
15	Rogi	31° 30' 40.495" N	78° 13' 38.867" E
16	Saryo	31° 31' 54.637" N	78° 16' 9.438" E
17	Shuda Rang	31° 31' 21.837" N	78° 15' 54.415" E
18	Tangling	31° 31' 19.712" N	78° 17' 7.965" E
19	Tarkhwa	31° 34' 3.173" N	78° 16' 4.413" E
20	Telingi	31° 33' 18.888" N	78° 16' 26.328" E
21	Thikru	31° 30' 44.238" N	78° 7' 42.833" E
22	Yusaring	31° 30' 32.284" N	78° 16' 21.420" E
23	Rarang	31° 32' 57.568" N	78° 0' 55.757" E
24	Kandhar	31° 34' 5.523" N	77° 59' 48.988" E
25	Panvi Khas	31° 32' 11.929" N	78° 0' 27.950" E
26	Nichar	31° 33' 13.078" N	77° 58' 32.358" E
27	Agade	31° 33' 13.868" N	77° 59' 20.813" E
28	Linge	31° 33' 32.904" N	77° 59' 3.865" E

29	Garade	31° 33' 32.304" N	77° 58' 2.489" E
30	Puje	31° 33' 19.798" N	77° 57' 53.666" E
31	Saki Charang	31° 33' 38.702" N	77° 57' 4.987" E
32	Punaspa	31° 33' 32.672" N	77° 56' 48.540" E
33	Sungra	31° 33' 47.357" N	77° 56' 8.292" E
34	Kangosh	31° 33' 10.138" N	77° 55' 52.333" E
35	Solding	31° 32' 49.579" N	77° 55' 5.747" E
36	Paunda	31° 33' 0.073" N	77° 55' 13.199" E
37	Palingi	31° 33' 14.382" N	77° 55' 31.923" E
38	Tranda	31° 33' 29.850" N	77° 53' 51.657" E
39	Karape	31° 33' 34.214" N	77° 53' 24.905" E
40	Chhonda	31° 33' 6.225" N	77° 53' 7.773" E
41	Tikrang	31° 32' 56.095" N	77° 53' 1.134" E
42	Nanaspo	31° 33' 15.429" N	77° 53' 1.010" E
43	Nigulsari	31° 33' 25.962" N	77° 53' 0.517" E
44	Thach	31° 33' 17.986" N	77° 52' 12.072" E
45	Pashpa	31° 34' 53.053" N	77° 50' 56.715" E
46	Busrag	31° 34' 46.551" N	77° 51' 35.674" E
47	Tapri	31° 31' 2.615" N	78° 5' 54.575" E
48	Punag Khas	31° 30' 38.601" N	78° 5' 47.375" E
49	Kutano	31° 31' 17.046" N	78° 7' 28.523" E
50	Jani	31° 30' 53.783" N	78° 4' 34.222" E
51	Burcha	31° 32' 26.628" N	78° 0' 9.993" E
52	Joktarang	31° 32' 3.479" N	78° 0' 44.108" E
53	Choling	31° 31' 15.974" N	78° 8' 23.466" E
54	Kafaur	31° 33' 57.205" N	77° 50' 32.090" E
55	Shilani	31° 33' 42.454" N	77° 50' 50.285" E
56	Chaura	31° 34' 12.727" N	77° 51' 11.614" E
57	Kagsthal	31° 32' 17.377" N	78° 1' 16.301" E
58	Yashang	31° 31' 41.325" N	78° 5' 0.385" E
59	Paring Khor	31° 32' 48.505" N	78° 0' 11.340" E
60	Up Mohal Khabo	31° 47' 55.028" N	78° 38' 36.678" E
61	Up Mohal Dubbling	31° 44' 39.870" N	78° 37' 51.358" E
62	Up Mohal Dabling	31° 44' 41.352" N	78° 36' 58.326" E
63	Up Mohal Yangti Kanda	31° 45' 9.597" N	78° 37' 53.650" E
64	Up Mohal Pooh	31° 45' 53.822" N	78° 35' 33.347" E
65	Up Mohal Labrang Khas	31° 45' 35.921" N	78° 35' 3.346" E
66	Up Mohal Danmochhe	31° 45' 15.602" N	78° 34' 32.322" E
67	Up Mohal Spiloo Khas	31° 39' 36.418" N	78° 26' 30.268" E
68	Up Mohal Kanam Khas	31° 40' 36.513" N	78° 27' 4.996" E
69	Namaklang	31° 40' 13.549" N	78° 26' 36.501" E
70	Up Mohal Foche	31° 40' 5.537" N	78° 27' 3.951" E

71	Up Mohal Kanam Nichla	31° 40' 24.925" N	78° 27' 5.689" E
72	Up Mohal Gyamil	31° 41' 10.723" N	78° 28' 58.039" E
73	Nakal Kuwa	31° 38' 24.699" N	78° 26' 44.485" E
74	Jangi Khas	31° 36' 41.060" N	78° 25' 42.658" E
75	Gramang	31° 36' 19.532" N	78° 26' 46.114" E
76	Khokpa	31° 35' 26.775" N	78° 26' 50.226" E
77	Samodayan	31° 34' 42.354" N	78° 26' 42.310" E
78	Shiling Khas	31° 35' 55.339" N	78° 26' 55.938" E
79	Akpa Khas	31° 35' 13.690" N	78° 23' 13.457" E
80	Khadura	31° 35' 31.019" N	78° 22' 21.141" E
81	Rarang Khas	31° 36' 5.917" N	78° 21' 8.247" E
82	Swaden	31° 36' 33.692" N	78° 19' 42.154" E
83	Sheelapur	31° 36' 44.687" N	78° 19' 25.562" E
84	Rispa Khas	31° 34' 36.836" N	78° 25' 23.437" E
85	Lijing	31° 34' 29.662" N	78° 25' 31.712" E
86	Holdang	31° 34' 45.563" N	78° 24' 14.786" E
87	Skiba	31° 34' 45.189" N	78° 22' 45.801" E
88	Kurpo	31° 34' 37.391" N	78° 22' 21.947" E
89	Kutian	31° 34' 43.213" N	78° 22' 5.666" E
90	Ribba Khas	31° 35' 16.790" N	78° 21' 51.263" E
91	Parga	31° 35' 27.770" N	78° 21' 28.906" E